Environmental Engineering

Unit I: Water Supply

Syllabus

Estimation of surface and subsurface water resources - Predicting demand for water - Impurities of water and their significance - Physical, chemical and bacteriological analysis - Waterborne diseases - Standards for potable water - Intake of water: pumping and gravity schemes.

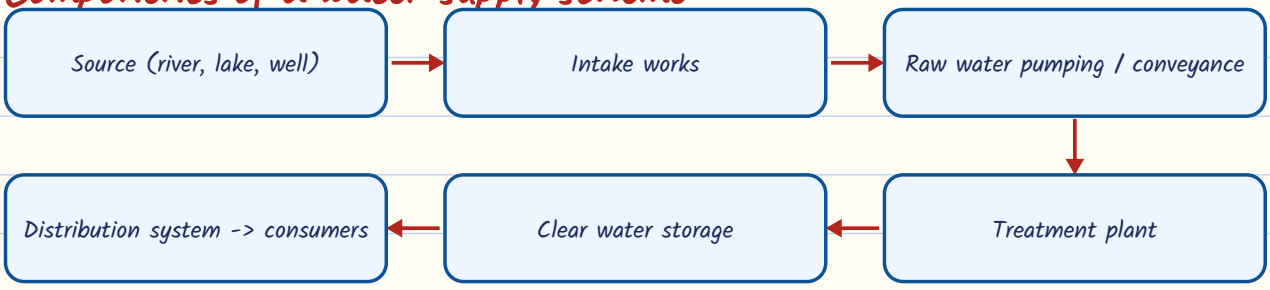
1. Introduction to Water Supply Engineering

Water supply engineering deals with collecting water from a source, treating it to make it safe, and delivering it to consumers in the **required quantity**, at **required pressure** and of **required quality**, at a reasonable cost.

Objectives of a water supply scheme

- To supply **safe and wholesome** (potable) water to the public
- To supply adequate quantity for domestic, industrial, public and fire-fighting needs
- To make water available at **all times** (continuous supply) and with sufficient pressure
- To keep the cost of construction, operation and maintenance **economical**
- To protect the public from waterborne diseases

Components of a water supply scheme



Types of water demand / needs

- **Domestic** - drinking, cooking, bathing, washing, flushing
- **Industrial** - cooling, boilers, process water
- **Public / civic** - road washing, parks, fountains, public toilets
- **Fire demand** - for fire fighting, needs high pressure and flow

- **Losses and wastes** - leakage, theft, meter errors (about 15%)

2. Sources of Water

All water on earth comes from rainfall (hydrologic cycle). Sources are classified into surface sources and subsurface (ground) sources.

Surface sources	Subsurface sources
Rivers, streams, lakes, ponds, impounding reservoirs	Wells, springs, infiltration galleries, infiltration wells
Water is **soft**, turbid, polluted, high bacteria	Water is **hard**, clear, low bacteria, more dissolved salts
Needs full treatment (sedimentation, filtration, disinfection)	Needs less treatment, often only disinfection
Large quantity available, variable with season	Quantity limited, depends on aquifer yield

Surface water sources

- **Rivers** - perennial rivers give large and continuous supply, but turbidity and pollution are high; intake works are needed
- **Lakes and ponds** - quiet water, suspended matter settles, but algae and weeds grow (taste and odour)
- **Impounding reservoirs** - a dam is built across a valley to store monsoon flow; used where rivers are non-perennial
- **Canals** - water is taken from irrigation canals; needs treatment

Subsurface (ground water) sources

- **Aquifer** - a water-bearing geological formation that stores and yields water (sand, gravel, fractured rock)
- **Aquiclude** - formation that stores but does not transmit water (clay)
- **Aquifuge** - neither stores nor transmits (solid granite)
- **Aquitard** - poorly permeable layer that transmits water slowly
- **Water table** - upper surface of the saturated zone; unconfined aquifer is under atmospheric pressure
- **Confined (artesian) aquifer** - bounded by impermeable layers, water is under pressure; artesian wells flow without pumping
- **Open (dug) wells** - shallow, large diameter, 5-10 m deep, get water from water table
- **Tube wells** - small diameter bore, 30-300 m; strainer tubes lower down

- **Springs** - natural outflow of ground water at surface
- **Infiltration galleries** - horizontal perforated pipes laid in water-bearing strata near rivers
- **Infiltration wells** - vertical wells sunk in the river bed or bank

3. Estimation of Surface Water Resources

The yield of a surface source depends on **rainfall**, catchment area and losses. For design of an impounding reservoir the **safe yield** is the maximum quantity available during the **driest period** (critical period).

Catchment area yield methods

- **Run-off** = Rainfall - (Evaporation + Infiltration + Transpiration + Depression storage)
- **Strange's table / curves** - run-off % based on catchment type (good, average, bad) and monsoon rainfall
- **Binnie's percentages** - run-off in terms of rainfall (based on Nagpur data)
- **Barlow's tables** - run-off coefficient (k) for different catchment classes: $Q = k \times P$
- **Inglis and De Souza formula** (for Western Ghats / Deccan): $R = 0.85\,P - 30.5$ (R, P in cm)

Rational formula (for small catchments)

$Q = C\,i\,A / 360$ (Q in m^3/s, i in mm/h, A in ha)

C = run-off coefficient (0.1 for forest to 0.9 for paved areas)

Storage capacity of reservoir (mass curve method)

- Plot cumulative inflow against time (**mass curve**)
- Draw tangents with slope equal to the demand rate (demand lines) from the peaks of the mass curve
- The maximum vertical intercept between the demand line and the mass curve gives the **required storage capacity**
- Reservoir capacity must also allow for evaporation, silting, and dead storage

4. Estimation of Subsurface (Ground) Water Resources

Darcy's law

Darcy's Law

$v = K\ i$ and $Q = K\ i\ A = K\ (dh/dL)\ A$

K = coefficient of permeability (m/day), i = hydraulic gradient, A = area of flow

Valid for laminar flow, Reynolds number less than 1 (up to about 10)

Specific yield and specific retention

- **Porosity (n)** = volume of voids / total volume
- **Specific yield (S_y)** = volume of water that drains by gravity / total volume of soil
- **Specific retention (S_r)** = water retained against gravity; $n = S_y + S_r$
- **Storage coefficient** = volume of water released per unit area per unit fall in head

Yield of wells

Steady flow into wells (Dupuit's theory)

Unconfined aquifer

$Q = pi\ K\ (h_2^2 - h_1^2)\ /\ ln(r_2/r_1)$

h_1, h_2 = water heights in observation wells at radial distances r_1, r_2

Confined aquifer

$Q = 2\ pi\ K\ b\ (h_2 - h_1)\ /\ ln(r_2/r_1)$

b = thickness of the confined aquifer

- **Specific capacity** of a well = Q / s (yield per unit drawdown)
- **Radius of influence** - distance from the well to where drawdown is zero
- **Cone of depression** - funnel shaped drop in water table around a pumped well
- Safe yield of a ground water basin = recharge - natural discharge

5. Population Forecasting and Water Demand

A water supply scheme is designed for a **design period** of 20-30 years (intake, treatment 15 years; pipes 30 years; pumps 10-15 years). So the future population must be predicted.

Methods of population forecasting

Method	Formula / idea	Used for
Arithmetic increase	$P_n = P + n \times$ (average increment per decade)	Old, large, stable cities
Geometric increase	$P_n = P (1 + r/100)^n$	Growing young cities
Incremental increase	$P_n = P + n \times X + n(n+1)/2 \times Y$ (X = avg increase, Y = avg of increments)	Average cities
Decreasing rate of growth	Rate of growth decreases with time, saturation limit assumed	Cities nearing saturation
Graphical method	Curve of past data extended / comparison with similar cities	Quick check
Logistic curve	$P = P_s / (1 + m\, e^{nx})$	Long term, S-shaped growth
Ratio / apportionment	Ratio of city to state population kept constant	Where state data known

Per capita demand (as per IS 1172 / CPHEEO)

Type of town	Water demand (lpcd)
Towns with piped supply, no sewerage	70
Towns with piped supply and sewerage	135
Metropolitan and mega cities	150 (CPHEEO manual up to 150-200)

Typical break-up of domestic demand (135 lpcd)

* Drinking 5, Cooking 5, Bathing 55, Washing clothes 20
* Washing house 10, Flushing of WC 30, Others 10
* Total 135 lpcd (plus 15% losses)

Fire demand formulae (Q in kilolitres/day, P in thousands)

- **Kuichling:** $Q = 3182 \sqrt{P}$
- **Freeman:** $Q = 1136 (P/5 + 10)$
- **National Board of Fire Underwriters:** $Q = 4637 \sqrt{P} (1 - 0.01 \sqrt{P})$
- **Buston:** $Q = 5663 \sqrt{P}$

Variations in demand

- **Maximum daily demand** = 1.8 x average daily demand
- **Peak hourly demand** = 1.5 x maximum daily demand = 2.7 x average
- Seasonal (summer high), daily (weekends), hourly (morning and evening peaks)
- Distribution system is designed for peak hourly demand; pumps for maximum daily demand

Factors affecting per capita demand

- Size of city, climate, standard of living, industrial and commercial activity
- Quality of water, pressure in mains, cost of water (metered supply cuts demand)
- Systems of supply (continuous or intermittent), sewerage facility, habits of people

Example (arithmetic method)

Populations: 1991 = 40,000; 2001 = 45,000; 2011 = 52,000. Find the 2031 population.

Average increment = (5000 + 7000)/2 = 6000 per decade

P_{2031} = 52,000 + 2 x 6000 = **64,000**

6. Impurities of Water and Their Significance

Natural water is never pure. It picks up impurities from air, soil and human activity. Impurities are classified as **suspended, colloidal** and **dissolved** (by particle size, >1 micron, 1 nm - 1 micron, <1 nm).

Impurity	Source	Significance
Suspended solids (silt, clay, algae)	Erosion, run-off	Turbidity, shields bacteria, removed by sedimentation and filtration
Colloids	Organic decay, clay	Colour and turbidity, need coagulation
Dissolved gases (O_2, CO_2, H_2S)	Air, decay	Corrosion, taste and odour
Calcium, magnesium	Rocks (limestone, dolomite)	Hardness, scale in boilers, soap wasted
Iron, manganese	Ground water, minerals	Red/black stains, bitter taste, bacterial growth
Chlorides	Sea water, sewage	Salty taste, indicates pollution
Sulphates	Gypsum, industrial waste	Laxative effect (above 400 mg/L), corrosion of concrete
Nitrates	Fertilizers, sewage	**Methaemoglobinemia (blue baby)** in infants
Fluoride	Fluorite-bearing rocks	Below 0.6 mg/L dental caries; above 1.5 mg/L **fluorosis**
Arsenic, lead, mercury, cadmium	Industry, geological	Toxic, carcinogenic, nervous system damage
Pathogens	Human and animal excreta	Waterborne diseases

7. Analysis of Water

A. Physical characteristics

- **Turbidity** - cloudiness due to suspended matter; measured by **nephelometer** in NTU; limit 1 NTU (acceptable), 5 NTU (permissible)
- **Colour** - true colour (dissolved) and apparent colour; measured by Tintometer/platinum-cobalt scale, limit 5 Hazen units (15 permissible)
- **Taste and odour** - from algae, decaying matter, chemicals; threshold odour number test
- **Temperature** - best 10-15 degrees C; affects chemical reactions, bacteria and dissolved oxygen
- **Electrical conductivity** - gives idea of total dissolved solids; **TDS** limit 500 mg/L (2000 permissible)
- **pH** - measured by pH meter; limit 6.5 to 8.5

B. Chemical characteristics

- **Alkalinity** - capacity to neutralize acids (HCO_3, CO_3, OH); titration with H_2SO_4 using phenolphthalein and methyl orange
- **Acidity** - capacity to neutralize bases (free CO_2, mineral acids)
- **Hardness** - due to Ca and Mg; **temporary** (carbonate, removed by boiling) and **permanent** (sulphate, chloride); EDTA titration with Eriochrome Black-T indicator; limit 200 mg/L (600 permissible)
- **Chlorides** - Mohr's method (silver nitrate titration, potassium chromate indicator); limit 250 mg/L
- **Dissolved oxygen (DO)** - Winkler's iodometric method
- **BOD** - oxygen used by bacteria to stabilize organic matter in 5 days at 20 degrees C
- **COD** - oxygen needed for chemical oxidation (dichromate reflux), COD > BOD
- **Residual chlorine** - orthotolidine (OT) test, DPD method
- **Nitrogen forms** - free ammonia (fresh pollution), albuminoid, nitrite (partly decomposed), nitrate (fully oxidised, old pollution)
- **Metals** - Fe, Mn, Pb, As by atomic absorption spectrophotometer (AAS)

Hardness classification

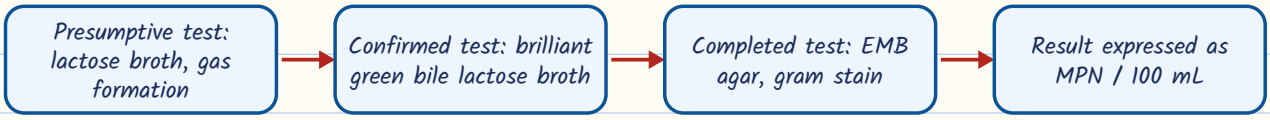
* 0 - 75 mg/L as $CaCO_3$: soft

* 75 - 150 : moderately hard
* 150 - 300 : hard
* above 300 : very hard

C. Bacteriological characteristics

Testing for every pathogen is difficult, so we test for **indicator organisms** - the coliform group (**E. coli**) which live in the human intestine and are present in large numbers in faeces. Presence of coliforms indicates faecal pollution.

- **Standard Plate Count** - total bacteria colonies on nutrient agar at 37 degrees C for 24 h, should be less than 500 per mL
- **MPN test (Most Probable Number)** - multiple tube fermentation: presumptive, confirmed and completed tests
- **Membrane filter technique** - sample filtered through 0.45 micron membrane, colonies counted on m-Endo media
- **Colilert / H_2S strip test** - quick field tests for presence/absence
- Drinking water standard: **zero coliform / E. coli per 100 mL**



8. Waterborne Diseases

Type	Disease	Causative agent
Bacterial	Cholera	Vibrio cholerae
Bacterial	Typhoid	Salmonella typhi
Bacterial	Bacillary dysentery	Shigella
Bacterial	Gastroenteritis	E. coli
Viral	Infectious hepatitis (Jaundice)	Hepatitis A / E virus
Viral	Poliomyelitis	Polio virus
Protozoal	Amoebic dysentery	Entamoeba histolytica
Protozoal	Giardiasis	Giardia lamblia
Helminthic	Guinea worm, round worm, tape worm	Dracunculus, Ascaris, Taenia
Water-based / vector	Malaria, dengue, filariasis (mosquitoes breed in water)	Parasites / viruses

- Spread by **faecal-oral route**: drinking contaminated water, food, flies
- Prevention: protected water source, treatment and disinfection, sanitary disposal of excreta, personal hygiene, chlorination, boiling of water

9. Standards for Potable Water

Potable water should be colourless, odourless, tasteless, free from pathogens, and have no harmful chemicals. Indian standards: **IS 10500:2012** (BIS) and the CPHEEO manual; also WHO guidelines.

Parameter	Acceptable	Permissible (no alternate source)
pH	6.5 – 8.5	No relaxation
Turbidity	1 NTU	5 NTU
Colour	5 Hazen	15 Hazen
TDS	500 mg/L	2000 mg/L
Total hardness	200 mg/L	600 mg/L
Chloride	250 mg/L	1000 mg/L
Sulphate	200 mg/L	400 mg/L
Fluoride	1.0 mg/L	1.5 mg/L
Iron	0.3 mg/L	No relaxation
Nitrate	45 mg/L	No relaxation
Arsenic	0.01 mg/L	0.05 mg/L
Residual chlorine	0.2 mg/L	1.0 mg/L
Coliform / E. coli	Absent in 100 mL	Absent

10. Intake of Water

An **intake** is a structure placed in the source to withdraw water and carry it by pipe or conduit to the treatment plant. A good intake gives water of the **best quality** at **all seasons**.

Factors for selecting the site of intake

- Should be **upstream** of the town and sewage outfalls, to get purest water
- Water available throughout the year, even in the driest period
- Located where the **river bed is stable** (not shifting), on the concave bank (outer side of a bend) where deep water is present

- Easy access in floods, safe against flood and ice, away from navigation channel
- Near the treatment plant to cut the cost of conveyance
- Free from floating matter, silt, and weeds

Types of intakes

Intake	Description	Suited to
Submerged intake	Pipe or chamber fully under water	Rivers and small lakes, economical
Wet intake (tower)	Masonry tower with openings (ports) at different levels, water enters the tower and is drawn by pump from a sump	Large rivers, deep lakes, reservoirs
Dry intake (tower)	Tower dry inside, water flows through ports to a conduit, valves inside tower	Deep, large lakes
Canal intake	Opening taken from a canal	Where canal water is used
Reservoir / dam intake	Draw-off at different levels through the dam body (sluice)	Impounding reservoirs
Floating / Jack well	Intake on a raft or well on bank with inlet pipes	Variable river level

- **River intake** - an intake well is built in bank or river, connected by pipes, with a **bell-mouth** and screen to prevent entry of fish and debris
- **Intake ports** are provided at various levels so water can be taken from the layer of best quality
- **Screens (coarse)** - bar screens to exclude floating matter; velocity through screen not more than 0.15 m/s
- Valves are provided to regulate and cut off flow

11. Conveyance of Raw Water: Gravity and Pumping Schemes

Water from intake is moved to the treatment plant, from there to service reservoirs, by one of the following systems.

Gravity system	Pumping system
Source is **higher** than the town, water flows by gravity	Source is **lower** than the town, pumps are used
Most economical and reliable, no power cost	Needs power, repair and maintenance, standby pumps

Gravity system	Pumping system
Needs pipes capable of carrying pressure; velocity 0.6 – 1.5 m/s	Pump may be directly to the distribution system or to elevated storage
Suitable for hill and spring sources	Typical for river and tube-well sources
Can be **gravity main** or **gravity + pumping combined**	Combined: pumping to a reservoir then gravity distribution is best

Pumping schemes

- **Direct pumping** – pumps feed the distribution system directly; no storage; pressure fluctuates; standby units needed; used for small towns
- **Pumping with elevated storage** – pumps work at uniform rate, excess goes to the service reservoir, and the reservoir meets the peak demand; most favoured
- **Pump types:** centrifugal pumps (most common), reciprocating, submersible (tube wells), air-lift pumps
- **Pump head** = static lift + friction losses + minor losses + residual head
- Power (kW) = (Q x H x gamma) / (eta x 1000) = 9.81 Q H / eta (Q in m^3/s, H in m, eta = overall efficiency)

Pipes for conveyance

- **Cast iron, ductile iron** – strong, long life (100 yrs), but heavy
- **Steel** – high pressure, light, welded; needs protection from corrosion
- **PVC / HDPE** – light, corrosion free, smooth, cheap, used for small diameters
- **RCC / prestressed concrete** – large diameter gravity mains
- **Asbestos cement** – now avoided because of health risk

Economical diameter of rising main

Lea's formula

$D = 1.22 \sqrt{Q}$ (D in m, Q in m^3/s)

Lamont / Marcel formula: D = 0.97 to 1.5 $\sqrt{Q}$, based on balancing pipe cost and pumping cost

12. Quick Revision Points

Remember for exam

* Design period: intake 30 yrs, pipes 30 yrs, pumps 15 yrs, treatment units 15 yrs
* Domestic demand 135 lpcd; max daily = 1.8 x avg; peak = 1.5 x max daily
* Geometric method for growing cities, arithmetic for old and saturated cities
* BOD uses 5 days at 20 degrees C; coliform standard is zero per 100 mL
* Fluoride limit 1.0 - 1.5 mg/L; nitrate 45 mg/L; TDS 500 mg/L; hardness 200 mg/L
* IS 10500:2012 is the Indian drinking water standard
* Intake is placed upstream of the town on the concave bank

Important questions

1. Explain the various sources of water supply with their merits and demerits.
2. Discuss the methods of forecasting population. Which is suitable for which city?
3. Explain the factors affecting per capita demand and the variations in demand.
4. Describe the physical, chemical and bacteriological tests for water quality.
5. What are waterborne diseases? Explain the preventive measures.
6. Explain IS 10500 drinking water standards for important parameters.
7. Describe the types of intake structures with neat sketches and the factors for selecting the site.
8. Compare gravity and pumping schemes of conveyance of water.
9. A city has population 25,000 (1991), 30,000 (2001), 38,000 (2011). Estimate the population in 2031 by the geometric and incremental increase methods.

Environmental Engineering

Unit II: Water Treatment

Syllabus

Objectives - Unit operations and processes - Principles, functions and design of water treatment plant units, aerators, flash mixers, coagulation and flocculation - Clariflocculator - Plate and tube settlers - Pulsator clarifier - Sand filters - Disinfection - Softening, removal of iron and manganese - Defluoridation - Desalination process - Residue management - Construction, operation and maintenance aspects.

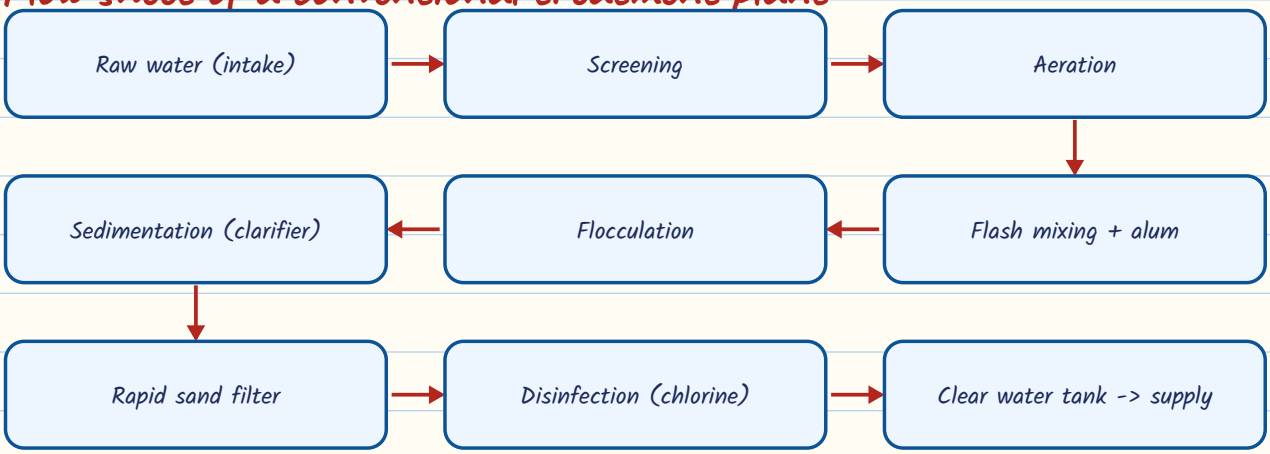
1. Objectives of Water Treatment

- To remove **suspended and colloidal impurities** (turbidity, colour)
- To remove **taste, odour** and dissolved gases
- To remove **pathogenic bacteria** and make water bacteriologically safe
- To reduce **hardness, iron, manganese, fluoride** and other harmful chemicals
- To make water **potable, palatable** and suitable for industries (non-corrosive, non-scale forming)

Unit operations and unit processes

- **Unit operations** - physical forces: screening, aeration, sedimentation, filtration, flotation
- **Unit processes** - chemical or biological reactions: coagulation, disinfection, softening, ion exchange, adsorption

Flow sheet of a conventional treatment plant



Choice of units depends on raw water quality. For a **clear ground water** only disinfection (and iron removal) may be enough; for a **turbid river water** the full flow sheet is needed.

2. Screening and Plain Sedimentation

- **Screens** are first unit; **coarse screens** (bar spacing 50-150 mm) remove floating debris, **fine screens** (6-12 mm) remove smaller matter; velocity through the screen below 0.8 m/s
- **Plain sedimentation** - settling of suspended particles under gravity without chemicals; removes sand and silt (above 0.02 mm), reduces load on later units
- Used as **pre-sedimentation** for very turbid rivers (turbidity above 500 NTU)

Theory of sedimentation

Stoke's law (settling velocity of a discrete particle)

$v_s = g (S_s - 1) d^2 / (18 nu)$

$g = 9.81 m/s^2$, S_s = specific gravity of particle, d = diameter, nu = kinematic viscosity

Valid for laminar flow, Reynolds number less than 1

- **Ideal settling basin zones:** inlet, settling, sludge, outlet
- **Surface overflow rate (SOR)** = $Q / A = v_0$; all particles with $v_s >= v_0$ are completely removed (independent of depth!)
- **Detention time** $t = V / Q$ = (volume of tank) / (flow rate)
- **Displacement efficiency** / flow-through period = length / velocity of flow

Design parameter	Value
Detention time (plain)	3 - 4 h
Detention time (with coagulation)	2 - 2.5 h
Surface overflow rate (plain)	15 - 30 $m^3/m^2/day$
SOR (with coagulant)	30 - 40 $m^3/m^2/day$
Horizontal velocity	Not more than 0.3 m/min (0.5 cm/s)
Depth	2.5 - 5 m
Length : Width	3:1 to 5:1 (rectangular); diameter up to 60 m (circular)
Weir loading	Below 300 $m^3/m/day$

Types of sedimentation tanks

- **Rectangular** - horizontal flow, length 3-5 times width, sludge scraped to a hopper
- **Circular** - radial flow, inlet at centre, outlet at periphery, rotating scraper
- **Upward flow tank** - used with sludge blanket
- **Continuous flow** vs **fill and draw** (intermittent) tanks

Design example

Q = 6 MLD, detention time 3 h. Volume = $6 \times 10^6/1000 \times 3/24 = 750\ m^3$.

Depth 3.5 m gives area = $214\ m^2$; take L : B = 4 : 1, B = 7.3 m, L = 29.3 m.

3. Aeration and Aerators

Aeration brings water and air into intimate contact to add oxygen and remove dissolved gases (CO_2, H_2S) and volatile compounds that cause taste and odour; it also oxidises **iron and manganese**.

Types of aerators

Aerator	Working
Gravity (cascade) aerator	Water falls over steps, trays or weirs; simple, no power
Spray aerator	Water sprayed through nozzles into air; 1.5-2 m head needed
Diffused air (bubble) aerator	Compressed air through porous plates at tank bottom
Mechanical aerator	Rotating paddles or turbines; surface agitation
Tray aerator	Perforated trays filled with coke or gravel to increase contact

- Contact time: 10-30 min in diffused aerators; air requirement 0.3 - 0.8 m^3 air per m^3 water
- The **flash mixer** is also called a rapid mixing unit (see next section)

4. Coagulation and Flocculation

Colloidal particles (0.001 - 1 micron) carry **negative charge** and repel each other, so they do not settle. **Coagulation** neutralises this charge using chemicals (coagulants), forming micro-flocs. **Flocculation** is slow mixing which joins micro-flocs into large settleable flocs.

Common coagulants

Coagulant	Formula	Notes
Alum (aluminium sulphate)	$Al_2(SO_4)_3.18H_2O$	Most common, cheap, pH 6.5-8.5, dose 10-50 mg/L
Ferrous sulphate (copperas)	$FeSO_4.7H_2O$	Needs lime; pH above 8.5
Ferric chloride	$FeCl_3$	Works at wide pH 4-11
Sodium aluminate	$NaAlO_2$	Used for coloured, soft waters
Poly aluminium chloride (PAC)	$Al_n(OH)_mCl$	Efficient, low sludge
Polyelectrolytes	Organic polymers	Coagulant aids, strengthen flocs

Reaction of alum with alkalinity

$Al_2(SO_4)_3 + 3Ca(HCO_3)_2 \rightarrow 2Al(OH)_3 + 3CaSO_4 + 6CO_2$

$Al(OH)_3$ is a gelatinous precipitate that traps colloids (**sweep floc**). If natural alkalinity is low, add lime or soda ash.

1 mg/L alum needs about 0.5 mg/L alkalinity as $CaCO_3$.

- **Optimum dose** found by **Jar test** - six beakers, different doses, rapid mix 1 min at 100 rpm, slow mix 20 min at 30 rpm, settle 30 min; the dose giving the clearest water is the optimum
- Factors: turbidity, pH, temperature, alkalinity, mixing intensity
- **Coagulant aids:** lime, activated silica, bentonite clay, polymers

Flash (rapid) mixer

- Provides violent mixing to disperse coagulant quickly: detention time **30 - 60 seconds**; velocity gradient $G = 300 - 1000\ s^{-1}$
- Types: **hydraulic** (weirs, hydraulic jump, Parshall flume, baffles), **mechanical** (propeller or turbine mixers)
- Mixing power: $P = G^2\ mu\ V$ (mu = dynamic viscosity, V = volume)

Flocculator

- Gentle stirring, $G = 10 - 75\ s^{-1}$, detention time **20 - 40 min**; Gt = 20,000 - 1,00,000 (dimensionless)
- Types: **mechanical** (paddle flocculators, slow-speed paddles with peripheral velocity 0.2-0.6 m/s) and **hydraulic** (baffle-wall: horizontal / over-and-under baffles with velocity 0.15-0.3 m/s)

- Water velocity must stay below 0.3 m/s in the outlet channel so that floc does not break



5. Clariflocculator

A **clariflocculator** combines the flocculation and sedimentation in one circular structure: a central flocculation zone surrounded by an outer clarification zone. It saves space and cost.

- Water with coagulant enters the **inner circular flocculation chamber** (mechanical paddle stirrers, detention 20-40 min)
- Flows under a skirt wall to the **outer annular clarifier** with slow upward/radial flow
- Settled sludge scraped to the central sump by a rotating bridge scraper
- Clarified water collected by peripheral launder over weirs
- Typical: flocculator detention 20-30 min, clarifier 2 h, SOR 30-40 $m^3/m^2/day$
- Used for large plants, with high turbidity raw water

6. Plate and Tube Settlers

These are **high rate settlers**: inclined plates or tubes are placed inside the settling basin to increase the **effective settling area** without increasing the tank area. Settling distance becomes very small (5 cm) so particles settle quickly.

- **Tube settlers** - square, hexagonal or circular tubes 25-50 mm diameter, inclined at **60 degrees** to horizontal; flow upward; sludge slides down the tubes by gravity
- **Plate settlers (lamella)** - parallel plates 50-100 mm apart, at 55-60 degrees; plates of PVC or FRP
- Reduce detention time from 2-3 h to **15-30 min**; loading rate up to 3-4 times conventional
- Used to upgrade overloaded old plants

Advantages and limits

* Compact, 40-70% area saving; good removal of fine flocs
* Needs good pre-flocculation, periodic cleaning (algae, sludge in tubes)
* Not for water with very high variations in turbidity

7. Pulsator Clarifier

A **sludge blanket (solids-contact)** clarifier with pulsating inflow. Raw water with coagulant is fed into a vacuum chamber; air is periodically released so that water is **pulsed** into the tank through perforated laterals at the bottom.

- Pulses keep the **sludge blanket** uniformly suspended and homogeneous (flocs formed in the blanket), providing very effective clarification
- Blanket height 1.5-2 m; excess sludge flows to concentrators
- Flat-bottom, square or rectangular; rise rate 1.5 - 3 m/h
- Good turbidity removal, but sensitive to sudden changes of flow, requires skilled operation

8. Filtration

Filtration removes remaining suspended and colloidal matter, bacteria and flocs when water passes through a porous bed (sand). It removes **90-99% of bacteria**, and cuts turbidity to below 1 NTU.

Feature	Slow sand filter	Rapid sand filter
Rate of filtration	100 - 200 L/h/m^2 (0.1-0.4 m/h)	3000 - 6000 L/h/m^2 (3-6 m/h)
Area required	Large	Small (1/30)
Sand effective size	0.2 - 0.4 mm, uniformity coeff. 2-3	0.45 - 0.7 mm, UC 1.3-1.7
Depth of sand	90 - 110 cm	60 - 75 cm
Pre-treatment	Plain sedimentation only	Coagulation + sedimentation needed
Mechanism	Biological action (schmutzdecke)	Mainly physical straining, adsorption
Cleaning	Scraping top 1-2 cm every 1-3 months	**Backwashing** every 24-48 h
Bacteria removal	98-99%	80-90%, so disinfection needed
Cost	High initial, low running	Low initial, high running

Rapid gravity filter - components

- **Filter bed** - 60-75 cm sand above 45 cm gravel in graded layers
- **Under-drainage system** - perforated laterals and manifold (central main), collects filtered water, distributes backwash
- **Filter box** - 3 to 4 m deep, RCC; area 10-100 m^2 each; at least 2 units
- **Wash-water troughs** above the bed
- **Rate controller** keeps rate constant, **loss of head gauge**
- **Backwashing** - reverse flow of clean water at 0.6 - 1 m/min to expand bed by 30-50%, for 5-10 min; used wash-water is 1-3% of filtered water; **air scour** helps
- Filter is washed when head loss reaches 2.5 - 3 m or effluent turbidity increases
- **Operational troubles:** air binding (negative pressure), mud balls, cracking of the bed, sand boils, filter clogging

Other filters

- **Pressure filters** - closed steel cylinders, rate 6-15 m/h, used in small plants, swimming pools, industry
- **Dual-media filters** - anthracite + sand; allow deeper penetration and higher rates
- **Multi-media filters, membrane filters (UF, MF), activated carbon filters**
- **Gravity filters designed:** Area = Q / rate; e.g., 6 MLD at 5 m/h needs 50 m^2

9. Disinfection

Disinfection means the **killing of pathogenic organisms** (not total sterilisation). It is the last treatment barrier. Ideal disinfectant: kills bacteria in short time, harmless to humans, leaves a **residual**, cheap, easy to apply and measure.

Methods

- **Chlorination** - most common; chlorine gas, bleaching powder (30-35% available Cl_2), sodium hypochlorite, chlorine dioxide
- **Ozonation** - strong, no taste, no residual; costly
- **Ultraviolet (UV) radiation** - 254 nm, no residual, needs clear water
- **Boiling** (household), **iodine, silver, potassium permanganate**

Chemistry of chlorination

$Cl_2 + H_2O \rightarrow HOCl + HCl$; $HOCl \leftrightarrow H^+ + OCl^-$

HOCl (**hypochlorous acid**) is the strong germicide (about 80 times more effective than OCl^-); it dominates at **pH below 7.5**

Free residual chlorine = $HOCl + OCl^-$; combined = chloramines (NH_2Cl, $NHCl_2$)

Chlorine demand, dose, residual

- **Chlorine demand** = dose applied - residual after contact time
- **Break-point chlorination** - chlorine is added until all ammonia and organics are oxidised; beyond the break point, the residual is **free** chlorine; ensures complete disinfection and removes taste and odour
- **Contact time** 30 min minimum, **residual chlorine** 0.2 mg/L at the consumer end (0.5 at the plant outlet)
- Dose of chlorine 0.2 - 0.5 mg/L for filtered water; bleaching powder dose = (chlorine dose x water volume) / (% available Cl_2)

Type of chlorination	Meaning
Plain / post-chlorination	Chlorine after filtration
Pre-chlorination	Before sedimentation, to oxidise and control algae
Super-chlorination	Large dose, then dechlorination with SO_2 or sodium thiosulphate
Dechlorination	Removing excess chlorine
Break-point chlorination	Dose beyond the break point

Disadvantage: formation of **trihalomethanes (THMs)**, which are carcinogenic, when chlorine reacts with organic matter. Chlorine applied by **chlorinators** (solution feed, vacuum type).

10. Water Softening

Hardness causes scale formation in boilers, wastage of soap, and is unsuitable for laundry and industry. Softening removes Ca and Mg ions. Needed when hardness exceeds 200 mg/L (up to 100 mg/L for industry).

Method	Principle
Lime-soda process	Lime removes carbonate hardness (temporary) and Mg; soda ash removes non-carbonate (permanent) hardness. Precipitates $CaCO_3$ and $Mg(OH)_2$
Zeolite (ion exchange) process	Water passes through sodium zeolite; Ca and Mg exchanged for Na. Regenerated with brine (NaCl)
Boiling	Removes temporary hardness only
Clark's method	Lime added to temporary hardness
Demineralisation	Cation and anion resin exchange, gives ion-free water

Method	Principle
Reverse osmosis / electrodialysis	Membrane separation

Lime-soda reactions

$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 + 2H_2O$

$MgSO_4 + Ca(OH)_2 \rightarrow Mg(OH)_2 + CaSO_4$; $CaSO_4 + Na_2CO_3 \rightarrow CaCO_3 + Na_2SO_4$

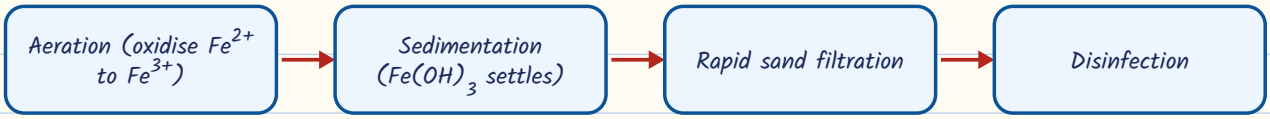
Zeolite: $Na_2Z + Ca(HCO_3)_2 \rightarrow CaZ + 2NaHCO_3$

Lime-soda vs zeolite

* Lime-soda: handles large hardness, low cost chemicals, sludge produced, residual hardness 30-50 mg/L
* Zeolite: no sludge, residual hardness near zero (0-5 mg/L), but only for clear water with low iron and turbidity, brine for regeneration

11. Removal of Iron and Manganese

- Present in ground water as dissolved **ferrous (Fe^{2+})** and **manganous (Mn^{2+})** forms with CO_2 and bicarbonate
- Problems: red-brown stains on clothes and fixtures, metallic taste, iron bacteria, clogging of pipes; limit Fe 0.3 mg/L, Mn 0.1 mg/L



- **Oxidation methods:** aeration, chlorine, potassium permanganate (**best for Mn**, since Mn needs pH above 9 for oxidation by air)
- **Lime softening** raises pH and precipitates Fe and Mn
- **Manganese zeolite / greensand filter** (oxidising bed regenerated with $KMnO_4$)
- **Ion exchange** - for low concentrations

12. Defluoridation

Fluoride above **1.5 mg/L** causes **dental and skeletal fluorosis** (mottled teeth, bone deformity). Common in ground water of Andhra Pradesh, Rajasthan, Gujarat, Tamil Nadu. Defluoridation brings the fluoride within 1 mg/L.

Method	Details
Nalgonda technique	Developed by NEERI. Alum + lime (or sodium aluminate) added, flash mixing, flocculation, settling, filtration. Domestic (bucket) and community plants. Alum dose 350-700 mg/L
Activated alumina adsorption	Water passes through activated alumina bed; regeneration with alum or NaOH
Bone charcoal / activated carbon	Adsorption, low cost
Ion exchange resin	Anion resin
Reverse osmosis, electrodialysis	High efficiency, costly
Other	Mixing with low-fluoride water, rainwater harvesting

13. Desalination

Removal of dissolved salts from **brackish** (TDS 1000-10,000 mg/L) or **sea water** (TDS about 35,000 mg/L) to make it fresh (TDS below 500).

Process	Principle
Reverse osmosis (RO)	Pressure (above osmotic pressure, 15-70 bar) pushes water through a semi-permeable membrane against natural osmotic flow; salt rejection 95-99%; energy 3-5 kWh/m^3
Electrodialysis (ED)	Ions move through ion-selective membranes under an electric field; for brackish water
Distillation (multi-stage flash, MED)	Water evaporated and condensed; heat energy; sea water
Solar still	Sun heat; small scale
Ion exchange	Resins swap ions; low salinity
Freezing	Ice crystals pure; less popular

- RO needs **pre-treatment** (filtration, anti-scalants, pH control) to prevent membrane fouling; waste **reject (brine)** 20-50% must be disposed safely
- **Nano-filtration** - removes hardness and some salts at lower pressure

14. Residue (Sludge) Management

Treatment plants produce sludge from sedimentation tanks, wash water from filters, and brine / reject from softening and RO. These must be handled carefully to avoid pollution.

- **Sources:** alum sludge ($Al(OH)_3$), lime sludge ($CaCO_3$), backwash water, iron sludge
- **Thickening** (gravity thickener) -> **conditioning** (polymers) -> **dewatering** (sand drying beds, filter press, centrifuge, belt press) -> **disposal**
- Lime sludge can be **recalcined** and reused, or used in soil conditioning
- Filter backwash is **recycled** to the head of the plant after equalisation
- Disposal: landfill, lagoons, discharge to sewer, brick and cement making, **RO reject** to evaporation ponds or marine outfall

15. Construction, Operation and Maintenance

- **Construction:** treatment units in RCC, water-tight, placed on hydraulic grade line so that flow is by gravity, with bypass lines and flow measurement (Parshall flume, venturi)
- **Operation:** daily record of raw and treated water quality, chemical doses by jar test, chlorine residual testing, flow rates, head loss
- **Maintenance:** cleaning of sedimentation tanks (desludging), backwashing of filters, painting, lubrication of scrapers and mixers, calibration of chlorinators and meters
- Standby units for pumps and chemical feeders; trained operators and safety (chlorine leak kits, gas masks)
- Safety: chlorine is toxic - store cylinders upright, in cool, ventilated rooms; leaks detected using ammonia swab

Remember for exam

* Alum dose 10-50 mg/L; flash mixer 30-60 s; flocculation 20-40 min; sedimentation 2-3 h
* RSF rate 3-6 m/h, SSF 0.1-0.4 m/h; backwash every 24-48 h
* Chlorine residual 0.2 mg/L, contact time 30 min; HOCl best at low pH
* Zeolite = no sludge; lime-soda = sludge; Nalgonda uses alum + lime
* RO is the principal desalination process

Important questions

1. Draw a flow sheet of a conventional treatment plant and explain the function of each unit.
2. Explain the theory of sedimentation and the design of a rectangular settling tank.
3. What is coagulation? Explain coagulants, jar test, flash mixing and flocculation.
4. Describe the working of a clariflocculator and tube settlers.

5. Compare slow and rapid sand filters. Explain the backwashing process.
6. Explain disinfection by chlorination and break-point chlorination.
7. Describe softening by lime-soda and zeolite processes.
8. Explain the removal of iron, manganese and fluoride from water.
9. Discuss desalination processes and management of residues from water treatment.

Environmental Engineering

Unit III: Water Storage and Distribution

Syllabus

Storage and balancing reservoirs - Types, location and capacity. Distribution system: layout, hydraulics of pipe lines, pipe fittings, valves including check and pressure reducing valves, meters, analysis of distribution systems, leak detection, maintenance of distribution systems, pumping stations and their operations - House service connections.

1. Storage and Balancing Reservoirs

Treated water is stored in **service (distribution) reservoirs** so that a uniform rate of treatment and pumping can meet the **fluctuating demand** of the town. They are placed between the treatment plant and the distribution system.

Functions of service reservoirs

- **Balance** the variation between supply (constant) and demand (variable)
- Provide a **reserve** for fire fighting and emergencies (breakdown of pumps or mains)
- Maintain **pressure** in the distribution system
- Allow pumps to work at a **uniform and efficient rate**, which cuts the cost of pumping
- Reduce the size of the **trunk main** (designed for average, not peak demand)
- Give time for **chlorine contact**

Types of reservoirs

Type	Description
Underground (ground level) reservoir	RCC / masonry tank at ground or below it; for flat towns, when natural elevation is not available; needs pumping to the system
Elevated reservoir (overhead tank)	Tank on a tower, for flat terrain; water flows by gravity; supports pressure; capacity small (up to a few lakh litres)
Surface (ground) reservoir on a hill	Natural high ground used, gravity supply
Standpipe	Tall cylindrical tank, upper part gives effective storage (balancing)
Balancing reservoir	Stores surplus during low demand and gives out during peaks
Storage reservoir	Large capacity, stores monsoon flow

Shapes and materials

- **Rectangular** - small capacity; **circular** - more economical in walls; **Intze type** - elevated tanks with conical dome bottom
- Materials: RCC (most common), steel (elevated), masonry, prestressed concrete, **GRP / HDPE** for small tanks
- Needs **roof cover**, ventilation, manhole, overflow, scour and outlet pipe; depth 3-6 m; **free board** 0.5 m, and floor slopes to a sump

Location and height of reservoirs

- Should be **centrally located** near the demand centre, at a high place to give gravity pressure
- Height decided to give the residual pressure of 7-15 m at the farthest point, with friction loss; **minimum 12-15 m above the ground** for a 3-storey building
- Elevated reservoir connected to the distribution through **inlet and outlet** (separate or same pipe)
- **Dead storage** (minimum level) never used

Capacity of a service reservoir

- Capacity = **balancing storage + breakdown storage + fire storage**
- **Balancing storage** is about 15-20% of the daily demand (or found from mass curve method)
- **Breakdown storage** - 1 / 4 of the total daily demand (for emergencies like pump failure)
- **Fire storage** - based on the fire demand for 4 hours (e.g., 1 lakh litres per hour for large towns)
- Total capacity is commonly **one third to one-half of the daily demand** (8 - 12 h)

Mass curve (analytical method) of balancing storage

Plot cumulative demand vs time (hours) → Draw supply line: cumulative supply (uniform rate) → Draw parallel lines tangent at highest and lowest points of the demand curve → Vertical distance between the tangents = **balancing capacity**

Example

Daily demand 4 MLD. Pump 24 h; max surplus = 8% and max deficit = 10% of daily demand. Balancing storage = (8 + 10)% x 4 = 0.72 ML. Add 25% breakdown (1 ML) + fire (say 0.2 ML) => total = **1.92 ML**.

2. Distribution System

The **distribution system** carries treated water from service reservoirs to the consumers in sufficient quantity, at **adequate pressure**, without deterioration of quality. It accounts for **40–70% of the cost** of the water supply scheme.

Requirements of a good distribution system

- Water to be supplied at all points with **required pressure** (min. 7–10 m, 12 m for 3-storey)
- Quantity adequate for domestic and **fire-fighting** demands
- **Water quality** should not deteriorate in pipes
- Easy **maintenance**, repair and extension; **water-tight** against leaks and entry of contaminants
- Layout with **minimum dead ends** and possible isolation of sections with valves

Methods of distribution

Method	Detail	Merits / Demerits
Gravity system	Source or reservoir above the town; no pumps	Cheapest, reliable. Suitable for hilly places
Pumping system	Direct pumping to mains	Pressure fluctuates, power failure stops supply
Combined (dual) system	Pumping to elevated tank, and gravity to the town; excess stored	**Most economical and reliable**

Systems of supply

- **Continuous system** – supply for 24 h, fresh water, good pressure for fire; but more wastage
- **Intermittent system** – supply for some hours (common in India), smaller pipes; but **risk of contamination** (negative pressure backflow), water stored in houses, pipe corrosion

3. Layout of Distribution Systems

Layout	Description	Advantages / Disadvantages
Dead end (tree) system	One main with sub-mains and branches; ends are dead	Simple, cheap, easy design; **stagnation at dead ends**, pressure drops if a part is repaired, poor fire flow
Grid iron system	Interconnected mains and branches in parallel (rectangular)	Free circulation, no dead ends, water available from many directions during repairs; **costly**, difficult analysis, many valves
Ring (circular) system	Main pipe round the town / area with branches	Good for fire; flow from both sides; needs more pipe length
Radial system	Zones from a central reservoir, flow radially outward	Quick, high pressure, simple design; for zoned towns

Choice of layout

* Depends on **town plan**, topography, street pattern and cost
* Grid iron or combined system for planned cities; dead-end for old and irregular towns
* Large cities use **zoning** (several pressure zones) from separate reservoirs

4. Hydraulics of Pipe Lines

Flow in pipes is turbulent. Head loss due to friction is found by **empirical formulae**; the pipe is designed for the required discharge and pressure.

Head loss formulae

Darcy-Weisbach: $h_f = f\,L\,v^2 / (2\,g\,D)$ (f = friction factor = 64/Re for laminar flow)

Hazen-Williams: $v = 0.849\,C\,R^{0.63}\,S^{0.54}$; $h_f = 10.67\,L\,Q^{1.852} / (C^{1.852}\,D^{4.87})$ [C = 100-140]

Manning's: $v = (1/n)\,R^{2/3}\,S^{1/2}$ (for large pipes flowing full: $R = D/4$)

- Common C values: new cast iron 130, old CI 100, PVC / plastic 150, concrete 120, steel 120
- **Minor losses** - entry, exit, bends, valves = $K\,v^2/2g$, small in long pipes
- **Velocity in mains** 0.6 - 2.0 m/s (minimum to prevent silting, maximum to limit loss and water hammer)
- **Pipes in series** - same Q, losses add; **pipes in parallel** - same head loss, flow splits
- **Equivalent pipe** - Dupuit formula: $L/D^5 = L_1/D_1^5 + L_2/D_2^5 + \ldots$

- **Water hammer** - sudden pressure rise on rapid closure of valve; Joukowsky: delta p = rho c v; prevent by slow valves, surge tanks, air vessels, relief valves

Pipe materials for distribution

Material	Merits	Demerits
Cast iron / ductile iron	Strong, long life (100 yrs), easy to join and tap	Heavy, tuberculation, corrodes
Mild steel / GI	Light, high pressure, easy to weld	Rusting, needs coating; GI small diameter (service)
PVC / uPVC / HDPE	Light, smooth, corrosion free, cheap	Low strength, affected by heat, UV
RCC / PSC	Large diameter, durable, low friction	Heavy, difficult to repair
Asbestos cement	Light, cheap	Brittle, health hazard (not used now)
Copper / lead	House service	Costly (lead is toxic)

5. Pipe Fittings

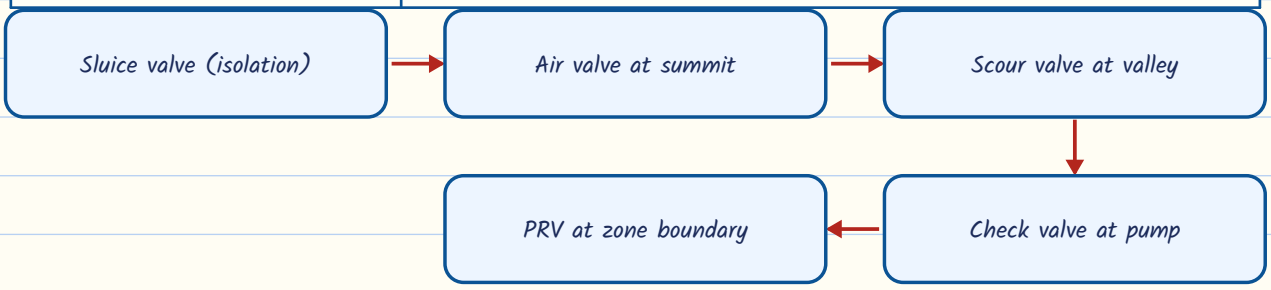
- **Sockets / couplings** - join two straight pipes
- **Bends, elbows (90, 45 degrees), tees (T), crosses, reducers / increasers** - change direction or diameter
- **Unions, flanges, nipples** - for easy removal
- **Plugs and caps** - closing ends; **bell-mouth**, **bib cocks**, **stop cocks**
- **Thrust blocks** (concrete anchor) at bends and tees to take hydraulic thrust
- **Expansion joints and flexible joints** (to take movement) and **specials**

Types of pipe joints

Joint	Used for
Spigot and socket (bell and spigot) - lead / rubber ring	CI pipes, most common
Flanged joint - bolts and gasket	Pumping stations, valve connections
Flexible (tyton) joint with rubber gasket	DI pipes, takes ground movement
Expansion joint	Long pipes in temperature change
Solvent cement joint	PVC pipes
Welded joint	Steel pipes

6. Valves

Valve	Function
Sluice (gate) valve	Isolate sections, open or close fully; installed at every 150-300 m and each branch
Check (reflux / non-return) valve	Allows flow in **one direction only**, prevents backflow; fixed on pump delivery, to protect the pump from reverse flow and water hammer
Pressure reducing valve (PRV)	Reduces and keeps a **constant downstream pressure** irrespective of upstream; for pressure zones; avoids bursts and leakage
Air relief (air) valve	Releases air accumulated at **summits** (high points) and admits air during emptying
Scour (blow-off) valve	At low points; drains the main and removes silt
Fire hydrant	Outlet for firefighting, at 100-150 m spacing; pressure 1-1.5 kg/cm^2
Altitude valve	Closes inlet when the elevated tank is full
Butterfly valve	Large mains, quick control, compact
Float valve, pressure relief valve	Controls water level / relieves excess pressure



7. Water Meters

Water meters measure the quantity of water consumed. They help in revenue collection, **reduce wastage** (up to 20-30%) and are used for finding leaks and for analysing the demand.

Type	Working
Positive displacement (volumetric) meter	Measures volume in each cycle (piston, oscillating disc); accurate at low flow; domestic
Velocity (inferential) meter	Turbine / propeller rotates with speed of flow, for large flows; (Woltmann type)
Electromagnetic / ultrasonic	No moving parts, accurate; used in large mains
Venturi meter / orifice meter	Pressure difference gives discharge; used in treatment plants and mains

Type	Working
Domestic meter / bulk meter	Small (15 mm) vs. large mains meter

- Meters need **strainer, regular testing and calibration**; installed in a meter chamber at a convenient location
- Advantages: equitable tariff, detects leakage, **discourages waste**; disadvantages: cost, maintenance, errors at low flow

8. Analysis of Distribution Systems

Analysis finds the **flows in the pipes and pressures at the nodes** of a network for a given demand. It is done to design pipe sizes and check the residual pressures.

Methods

Method	Use
Equivalent pipe method	Replace a part of network by single equivalent pipe, simple
Circle method	For a circular (ring) layout
Electrical analogy	Analog computer (Mc Ilroy analyser)
Hardy Cross method	Most common, iterative, for loops
Newton-Raphson, linear theory, computer packages	EPANET, WaterGEMS, LOOP, for large networks

Hardy Cross method (method of balancing heads)

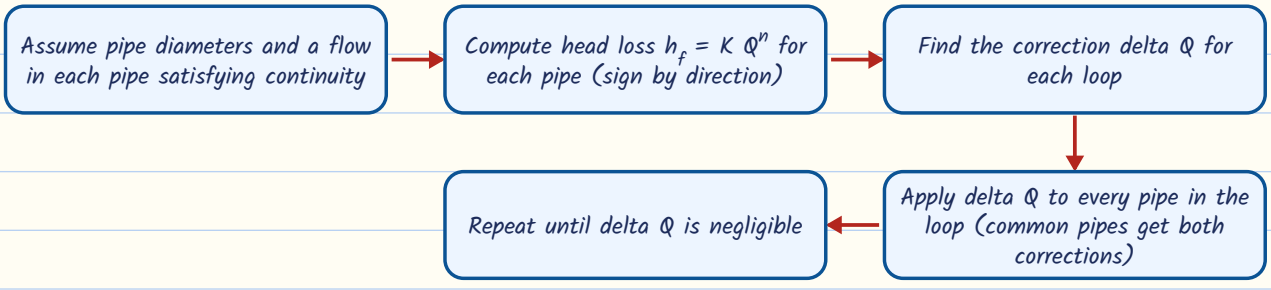
Principles

1. At every node, **inflow = outflow** (continuity): sum of $Q = 0$

2. In every closed loop, **algebraic sum of head losses = 0**: sum of $h_f = 0$ (clockwise +ve, anticlockwise -ve)

$h_f = K Q^n$; $n = 1.85$ (Hazen-Williams) or 2 (Darcy-Weisbach)

Correction: $\Delta Q = -\text{sum}(K Q^n) / \text{sum}(n K Q^{n-1}) = -(\text{sum } h_f) / (n \text{ sum}(h_f/Q))$



- Calculated pressure at nodes = elevation of reservoir − head loss − ground level; **residual pressure** at critical node must be at least 7 – 15 m
- **Design for peak demand** (maximum hourly) and check for maximum daily demand + fire flow
- Maximum **fire demand condition**: critical point gets the fire flow plus normal demand

9. Leak Detection and Water Losses

Unaccounted-for-water (UFW) in Indian cities is 30–50%. Losses occur from leaking pipes, joints, valves, house connections, illegal taps, and meter errors.

Methods of leak detection

- **Water audit** and **district metering areas (DMA)** – measure inflow in a district at night (minimum night flow) to detect leaks
- **Sounding (listening) rod / geophone** – hear the hissing sound of leak on valves and hydrants
- **Leak noise correlators** – two sensors on pipe, find the leak position from time delay
- **Ground microphones, acoustic loggers, tracer gas (helium, hydrogen), infrared thermography, ground-penetrating radar**
- **Pressure testing** of pipelines, **visual observation** of wet patches or fall in pressure
- **Control of leakage:** pressure management using PRVs, repair of leaks at once, replacement of old pipes, good quality fittings, regular meter calibration and quick service-connection repair
- **Active leakage control** programme with trained teams, **GIS and SCADA** for network monitoring

10. Maintenance of Distribution Systems

- **Periodic flushing** of mains by opening scour valves and hydrants, to remove silt and sediment
- **Swabbing and cleaning** of pipes (pigging) to remove tuberculation, followed by cement-mortar lining
- **Disinfection of new and repaired mains** with chlorine (50 mg/L for 24 h, or 20 mg/L for 24 h) before commissioning
- **Valve exercising and inspection** every year; hydrants maintained

- Maintain **residual chlorine** (0.2 mg/L at the end); sample water at various points to check bacteriology
- Prevention of **cross connection** (sewer and water pipes: water must be above sewer with 3 m clear distance)
- **Corrosion control:** lining, pH adjustment, cathodic protection
- **Records, maps** and GIS of the pipe network, repair crews, and public complaint system

11. Pumping Stations and Their Operations

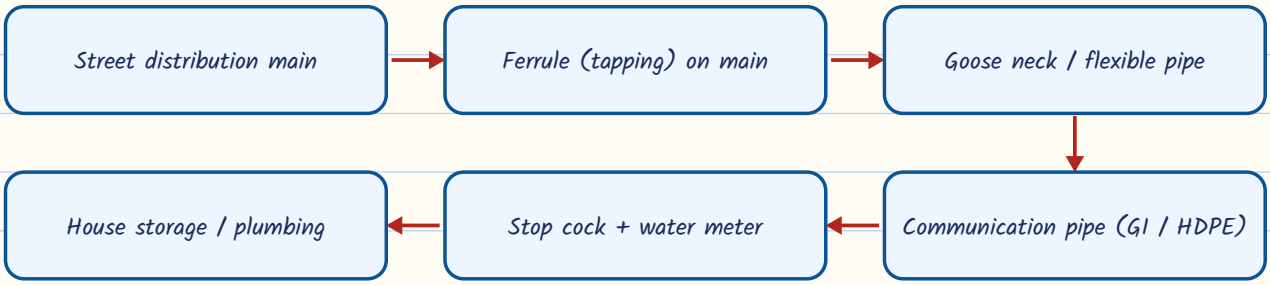
Pumping stations lift water from the source (raw water), through the treatment plant (intermediate), and to the distribution system (**booster** and **high-lift pumping**).

Pump type	Notes
Centrifugal pump	Most common; impeller, volute casing; high discharge, smooth flow; needs priming (except self-priming and submersible)
Reciprocating pump	Piston; high head, low discharge; pulsating flow
Submersible pump	Motor and pump together under water, for deep tubewells
Axial and mixed flow pump	Low head and high discharge
Jet pumps, air lift pumps, hydraulic ram	Special

- **Components:** suction well (sump), pumps and motors, delivery pipes, **foot valve and strainer**, check valve, sluice valve, priming device, control panel, **surge protection**, standby units (one standby per 3-4 working pumps)
- **Operation:** priming, start with delivery valve closed (for centrifugal), then open slowly; record discharge, head, current, voltage, bearing temperature and vibration; **log book**
- **Maintenance:** lubrication, replacement of seals and bearings, alignment, cleaning of strainer, painting, testing of motors, **preventive maintenance schedule**
- **Pump efficiency** = water power / input power; Water power (kW) = gamma Q H / 1000, 70-85% overall efficiency
- **Selection** by head-discharge curve, NPSH (available > required, to avoid **cavitation**), speed and cost

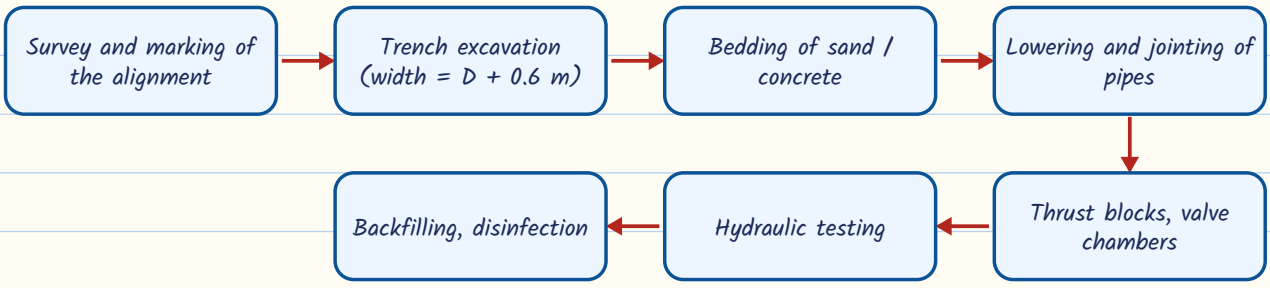
12. House Service Connections

A **service connection** is the pipe arrangement that connects the **street main** with the house plumbing.



- **Ferrule** is screwed into the main by a tapping machine under pressure
- **Goose neck** (lead, copper or PE) gives flexibility against settlement
- **Communication pipe** - from main to the property boundary, up to the stop cock; **service pipe** - inside the premises
- Diameter 12 - 25 mm; **stop cock** in a pit near the boundary; **meter** installed after the stop cock
- Connection pipe should not be near drains; avoid **back siphonage** (use non-return valves, break tank) to prevent contamination of the main
- Material: GI, MDPE/HDPE, copper (earlier lead is banned)

13. Laying and Testing of Pipes



- **Depth of cover** - minimum 1 m (0.9-1.2 m) below the road to avoid damage from traffic loads and frost
- **Water mains are laid above sewers**, with minimum 3 m horizontal separation, and on the **opposite side** of the street from drains where possible

- Pipes laid in straight lines and uniform gradients, with **air valves at summits and scour valves at valleys**
- **Hydrostatic test** - fill the section with water, pressurise to **1.5 times the working pressure** (or as specified, usually 2-3 kg/cm^2 above), hold for 1-2 h; leakage should not exceed permissible limits
- **Bacteriological test** after disinfection; flush before commissioning

Corrosion of pipes and its control

- **Causes:** acidic or soft water, dissolved oxygen, differential aeration, galvanic action between dissimilar metals, stray currents, bacteria (sulphate reducing, iron)
- **Tuberculation** - build-up of rust nodules, reduces the carrying capacity (C value falls from 130 to 90)
- **Control:** cement-mortar or epoxy lining, bitumen coating, **pH and alkalinity control** (lime, soda ash), **corrosion inhibitors** (polyphosphates, silicates), **cathodic protection** (sacrificial anodes), avoid dissimilar metals

14. Design Problems (solved examples)

Example 1: Capacity of an elevated tank

Population 30,000 at 135 lpcd => demand = 4.05 ML/day. If pumped only 8 h/day, the pumping rate = 4.05/8 = 0.506 ML/h (continuous 24 h pumping gives 0.169 ML/h).

Take balancing storage = 33% of daily demand = 1.34 ML.

Add fire storage 0.2 ML => total capacity = **1.54 ML** (about 1.6 ML).

Diameter of circular tank (depth 4 m): A = 1540/4 = 385 m^2, so D = 22.1 m.

Example 2: Head loss in a pipe (Hazen-Williams)

L = 1000 m, D = 300 mm, Q = 0.1 m^3/s, C = 100.

$h_f = 10.67 \times 1000 \times (0.1)^{1.852} / (100^{1.852} \times 0.3^{4.87})$

$= 10.67 \times 1000 \times 0.01406 / (5058 \times 0.00285) = 150 / 14.4 =$ **10.4 m**

Velocity v = Q/A = 0.1/0.0707 = 1.41 m/s (within 0.6-2 m/s, OK).

Example 3: Hardy Cross, first iteration

Loop ABCD: Q_{AB} = 40 L/s (K = 2.0), Q_{BC} = 15 (K = 3.0), Q_{AD} = 25 (K = 3.5), Q_{DC} = 10 (K = 2.5) in the clockwise sense (AB, BC clockwise; AD, DC anticlockwise).

$h_f = K Q^2$ (in suitable units): AB = +3.2, BC = +0.68, AD = -2.19, DC = -0.25 => sum h_f = 1.44

sum(2 K Q) = 0.475; delta Q = -1.44/0.475 = **-3.0 L/s** (reduce clockwise flows by 3.0, increase anticlockwise by 3.0) and repeat.

Pressure zones and distribution in hilly towns

- Pressure in a pipe should not exceed **60–70 m** (6–7 kg/cm^2) in town; high pressure leads to bursts, high leakage
- For tall differences in elevation, the town is divided into **zones**, with a separate reservoir or PRV for each; a **break pressure tank** is used in long gravity mains in hills
- **Dual supply / interconnection** between zones is provided for emergencies

Remember for exam

* Capacity of service reservoir = balancing + breakdown + fire storage
* Dual system (pumping + gravity) is the most economical; grid iron layout has no dead ends
* Hardy Cross: sum of Q at node = 0, sum of head losses in a loop = 0
* Hazen-Williams C: new CI = 130; PVC = 150; velocity 0.6 – 2 m/s
* Check valve = no backflow; PRV = constant downstream pressure; air valve at summits; scour valve at low points
* Hydrants at 100–150 m; minimum residual pressure 7–15 m

Important questions

1. Explain the types, location and capacity of service reservoirs. How is the balancing storage determined?
2. Describe the different layouts of distribution systems with merits and demerits.
3. Explain the methods of distribution: gravity, pumping and dual systems.
4. Explain the various valves used in distribution systems with their functions.
5. Describe the Hardy Cross method for analysing a pipe network.
6. Write notes on water meters and leak detection methods.
7. Discuss the maintenance of distribution systems.
8. Explain the working and operation of a pumping station.
9. Describe the house service connection with a neat sketch.

Environmental Engineering

Unit IV: Planning and Design of the Sewerage System

Syllabus

Characteristics and composition of sewage - Population equivalent - Sanitary sewage flow estimation - Sewer materials - Hydraulics of flow in sanitary sewers - Sewer design - Storm drainage - Storm runoff estimation - Sewer appurtenances - Corrosion in sewers - Prevention and control - Sewage pumping - Drainage in buildings - Plumbing systems for drainage.

1. Introduction to Sewerage

Sewage is the liquid waste from a community: domestic, industrial and storm water (with infiltration). **Sewerage** is the system of collecting, conveying, treating and disposing of it. **Sanitation** is the whole job of keeping the community clean and healthy.

Terms to know

Term	Meaning
Sewage / wastewater	Liquid waste from houses, institutions, industries
Sullage	Wastewater from bathrooms, kitchens, washing (without human excreta)
Sanitary sewage	Domestic + industrial sewage (with excreta)
Storm water	Run-off from rainfall
Sewer	Underground pipe carrying sewage
Sewerage	Collection system (sewers) with appurtenances
Night soil	Human excreta
Refuse / garbage	Solid waste

Systems of sewerage

System	Description	Merits / Demerits
Separate system	Sanitary sewage and storm water in **separate** sewers	Smaller sewers for sewage, cheaper treatment (no storm water), no overflow of sewage; **two sets of pipes cost more**, storm sewer lies idle in dry weather
Combined system	One sewer carries both	One pipe only, cheaper where rainfall is uniform, self-cleansing in rain; **large sewers, treatment overloaded in rains, overflow** problem
Partially separate (partially combined)	Part of storm water (roof and courtyards) goes into the sanitary sewer	Compromise, **good cleansing**

- Separate system is **preferred in India** (heavy monsoon rainfall in short duration)
- Combined system is suited where rainfall is evenly spread, or in areas with dense population and no space for two pipes

2. Characteristics and Composition of Sewage

Fresh sewage is **grey, turbid** with slight odour; **stale or septic** sewage is dark, with a foul smell of H_2S. Composition: **99.9% water** and only **0.1% solids** (suspended + dissolved). Of the solids about 70% is organic, 30% inorganic.

Physical characteristics

- **Temperature** - slightly higher than water supply (20-30 degrees C), affects biological activity
- **Colour** - grey (fresh), black (septic); **odour** - from H_2S, indole, skatole
- **Turbidity** - due to suspended and colloidal solids
- **Solids** - Total (TS), Suspended (SS), Dissolved (DS), **volatile (organic)** and **fixed (inorganic)**; settleable solids measured by Imhoff cone

Chemical characteristics

Parameter	Meaning / typical value (domestic sewage)
BOD_5	Oxygen needed by bacteria to oxidise organic matter in 5 days at 20 degrees C; **200 – 300 mg/L**
COD	Oxygen to oxidise organic matter chemically; 400 – 600 mg/L; COD > BOD
Dissolved oxygen (DO)	Nil in septic sewage; the discharge must keep DO > 4 mg/L in stream
Total suspended solids	200 – 400 mg/L
Nitrogen	Organic N, ammonia, nitrite, nitrate; TKN 40-60 mg/L; ammonia indicates fresh pollution
Phosphorus	5 – 15 mg/L, causes **eutrophication**
pH	6.5 – 8; **chlorides** 100 – 200 mg/L; **alkalinity** 100-200 mg/L
Oil and grease	Reduce oxygen transfer; remove in grease traps
Heavy metals and toxic compounds	From industries (Pb, Cr, Cd, Hg), toxic to bacteria

Biological characteristics

- Bacteria (aerobic, anaerobic, facultative) – **billions per mL**; viruses, protozoa, helminth eggs, algae, fungi
- **Coliform test (MPN)** indicates faecal contamination
- Bacteria decompose organic matter – the basis of **biological treatment**

BOD equation

$BOD_t = L_0 (1 - e^{-kt})$ [first-order reaction]

L_0 = ultimate BOD; k = BOD rate constant (0.23 per day at 20 degrees C for sewage, base e = 0.1 per day base 10)

Temperature correction: $k_T = k_{20} \times (1.047)^{T-20}$; BOD_5 is about 68% of ultimate BOD

3. Population Equivalent

To express the strength of industrial wastewater in terms of domestic sewage, the **population equivalent (PE)** is used. It is the population which would give the same BOD load as the industrial waste.

Population equivalent

PE = (BOD_5 of waste in kg/day) / (BOD per capita per day)

Per capita BOD load = **0.08 kg/day (80 g/day)** (some books use 0.045-0.085)

Also PE = (flow of waste in L/day x BOD mg/L) / (80 g/day), with units adjusted

Example: A dairy releases 500 m^3/day with BOD 600 mg/L => 300 kg/day => PE = 300/0.08 = **3750**

4. Estimation of Sanitary Sewage Flow

- About **80%** of water supplied becomes sewage (the rest is lost in lawn sprinkling, evaporation, leaks)
- **Average dry weather flow (DWF)** = 80% x per capita water supply x population = 0.8 x 135 = **108 lpcd**
- Add **infiltration** of ground water through joints: 5-10% of DWF (or 0.0005 to 0.005 m^3/s/km, or 1500-2000 L/day/km/cm diameter)
- **Industrial waste** added separately; **public places** also included

Flow	Multiplying factor
Average DWF	1
Maximum flow (peak)	**2 to 3 times** the average (depends on population)
Minimum flow	1/3 to 1/2 of the average (for population over 1 lakh, up to 1/2)
Design flow for sewers	**Peak flow**, 2.25 - 3 x average

Peak factor (CPHEEO)

Population up to 20,000 : 3.0; 20,000 - 50,000 : 2.5; 50,000 - 7.5 lakh : 2.25; above 7.5 lakh : 2.0

Or Harmon's formula: M = 1 + 14/(4 + sqrt(P)), P in thousands

5. Sewer Materials

Material	Merits	Demerits / use
Stoneware (vitrified clay)	Resists corrosion, smooth, cheap, small sizes, long life	Brittle, difficult to join; up to 300 mm
Cast iron / ductile iron	Strong, tight, for pumping mains, under bridges, high pressure	Costly, corrodes in acidic sewage
Concrete (plain, RCC, PSC)	Large diameter, any shape, strong, durable	Attacked by H_2S acid, heavy, joints; plain up to 600 mm
Brick masonry	Large trunk and storm sewers; any shape	Rough, porous, high friction
Asbestos cement	Light, long length	Not used now
Plastic (uPVC, HDPE, GRP)	Light, **smooth, corrosion free**, quick, leak-proof	Low strength, damage by heat; small-medium sizes
Steel	Pressure mains, bridges	Corrosion

- **Shapes of sewers:** circular (most common, max hydraulic efficiency, easy to make), egg-shaped (good self-cleansing at low flows), **horse-shoe / semi-elliptical / rectangular / U-shaped** (large trunk and storm drains)

6. Hydraulics of Flow in Sanitary Sewers

Sewers are designed to **flow partly full** under gravity (open channel flow) and carry maximum flow at depth near 0.8D. Flow in sewers follows Manning's, Chezy's or Hazen-Williams formula.

Flow formulae

Manning: $V = (1/n)\ R^{2/3}\ S^{1/2}$; $Q = A\ V$ (n = 0.012 - 0.015 for sewers)

Chezy: $V = C\ \sqrt{RS}$; **Kutter** and **Crimp & Bruge:** $V = 83.5\ R^{2/3}\ S^{1/2}$

Full flow circular pipe: $R = D/4$, so $V_{full} = (0.397/n)\ D^{2/3}\ S^{1/2}$

Self-cleansing velocity

- Velocity at which **sediments (grit) are not deposited**, and sewage solids are carried; **minimum 0.6 m/s** at the peak (some codes 0.8 m/s at design flow); **maximum 3 m/s** (non-scouring, avoid abrasion of the sewer); typical 0.8 - 2.4 m/s
- Shield's / Camp formula: $V = (1/n)\ R^{1/6}\ \sqrt{8\ \beta\ (S_s - 1)\ d / f}$ where β = 0.04-0.8, d = grit diameter

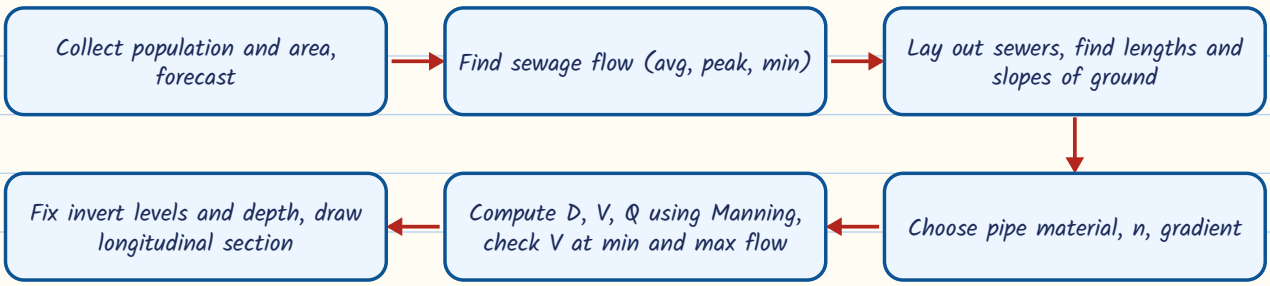
Partial flow in circular sewers (proportionate depth)

Depth ratio d/D	Q / Q_{full}	V / V_{full}
0.25	0.15	0.77
0.50	0.50	1.00
0.80	0.98	1.14 (max V at 0.81D)
0.94	1.07 (max Q near 0.94D)	1.10
1.00	1.00	1.00

Key points

* Max velocity occurs at 0.81 D, max discharge at 0.94 D
* Sewers are designed for 0.5 - 0.8 D at design flow, to leave room for gases and extra flow
* Velocity at minimum flow should be above 0.45 m/s (self-cleansing check)

7. Design of Sanitary Sewers



- **Design period** 30 years; sewers designed for **maximum flow running 0.8 full** (diameter D for peak with d/D = 0.8)
- **Minimum diameter:** 150 mm for public sewers (200 mm for CPHEEO in some cases), 100 mm for house connections
- **Minimum depth of cover:** 1 - 1.2 m (to protect from loads, avoid freezing, serve basements); **maximum depth** 6 m or less
- **Manholes** at changes of direction, gradient, size, junction, and every **30 m (up to 300 mm)**, 60 m for larger
- **Invert level** is the lowest point of the inside of the pipe; the flow line must keep downstream **crown** not above upstream crown **(match crown or match 0.8 depth lines)**

Example

Population 40,000 at 135 lpcd. Sewage = 0.8 x 135 x 40,000 = 4.32 MLD = 0.05 m^3/s. Peak factor 2.5 => Q = 0.125 m^3/s.

Slope 1 in 500 (S = 0.002), n = 0.013, running 0.8 full: Q_{full} = 0.125/0.98 = 0.1276 m^3/s.
Q_{full} = (0.312/n) $D^{8/3}$ $S^{1/2}$ => $D^{8/3}$ = 0.1276 x 0.013 / (0.312 x 0.0447) = 0.119 => D = 0.45 m; provide **450 mm pipe**. Check V_{full} = Q/A = 0.1276/0.159 = 0.80 m/s.

8. Storm Drainage and Storm Runoff Estimation

Storm drains carry the rain water from roofs, roads and open areas to natural streams, to prevent flooding. The design depends on **rainfall intensity, area, and the surface (imperviousness).**

Rational method

Q = (1/360) C i A (Q in m^3/s, i in mm/h, A in hectares)

or Q = C i A with i in m/s, A in m^2

C = run-off coefficient (impervious 0.8 - 0.95, roads 0.7-0.9, lawns 0.05-0.2, parks 0.1-0.25, residential 0.3-0.5)

- **Time of concentration (t_c) = inlet time** (overland flow into the first inlet, 5-20 min) + **time of flow** in the drain (length / velocity); the duration of the design storm equals t_c
- **Intensity-duration-frequency (IDF) formula:** i = a / (t + b)^n or i = K T^x / $(t + a)^n$; design **return period 2-10 years** for towns, up to 25-50 for important areas
- Methods for large areas: **hydrograph methods, Unit hydrograph, Kuichling, Burkli-Ziegler (i = ...), Mc Math**, SCS-CN method, computer models (SWMM)
- **Weighted C** for composite area: C = sum(C_i A_i) / sum(A_i)

Example

A 20 ha area: 40% roads and roofs (C = 0.85), 60% lawns (C = 0.15). Weighted C = 0.4 x 0.85 + 0.6 x 0.15 = 0.43.

t_c = 20 min, intensity 60 mm/h. Q = 0.43 x 60 x 20 / 360 = **1.43 m^3/s**

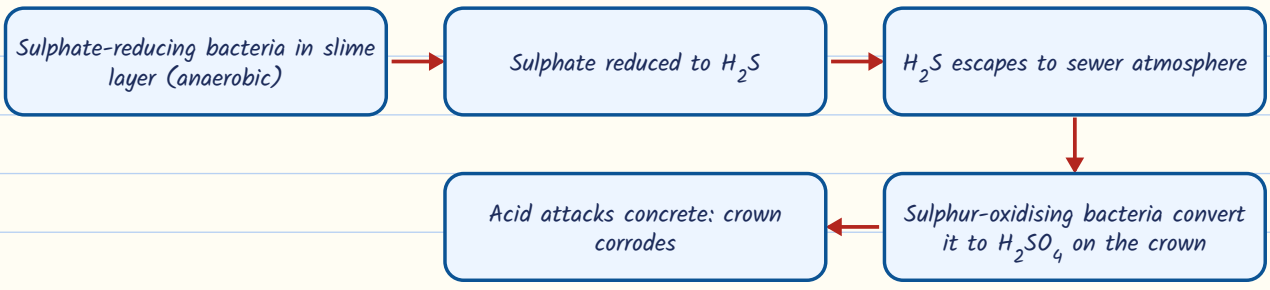
- Storm sewers are usually **open drains** along roads (lined, rectangular, trapezoidal or U-shaped), or **closed** large concrete pipes and box culverts
- **Inlets:** kerb inlets, gutter inlets, **catch basins (pits)** at 30-60 m to trap grit; **storm water overflow** weirs in combined system; **detention ponds, rain-water harvesting** to reduce the peak flow

9. Sewer Appurtenances

Appurtenance	Purpose
Manhole	Access for inspection and cleaning; at junctions, bends, change of size, every 30–90 m. Types: shallow (0.9 m), normal (1.5 m), **deep (above 1.5 m)** with steps and drop manhole. Cover CI or RCC
Drop manhole	Brings a high branch sewer down to a lower main, preventing splashing and erosion
Lamp hole	Straight sewers up to 300 mm, a lamp lowered to check alignment
Clean-out	Pipe up to the surface at the upper ends of sewers for flushing
Catch basin / catch pit	Retain grit, debris from the street inlets
Inlets (street, gully traps)	Admit surface storm water; grating prevents debris entering
Flushing tanks	Flush sewers at dead ends, avoids silting; automatic siphon
Grease and oil traps	Hotels, garages: remove grease
Inverted siphon (depressed sewer)	Crossing under a stream, road or railway; sewer dips below, runs under pressure; velocity above 1 m/s at the minimum flow
Storm overflow (regulator)	In combined system, to divert excess storm flow
Ventilating shafts	Remove sewer gases
Sewer ventilation	Prevents explosion and corrosion; via manhole covers, vent pipes

10. Corrosion in Sewers: Prevention and Control

Corrosion of concrete (and metal) sewers is chiefly due to **hydrogen sulphide (H_2S) formed in anaerobic septic sewage**.



Reactions

SO_4^{2-} + organic matter -> S^{2-} + CO_2 + H_2O (sulphate-reducing bacteria)

$H_2S + 2O_2 \rightarrow H_2SO_4$ (Thiobacillus, above the water line)

$H_2SO_4 + CaCO_3$ (cement) -> $CaSO_4 + H_2O + CO_2$ (expands, cracks, spalls)

- **Other problems of H_2S:** foul odour (rotten egg smell), toxic gas (**dangerous to maintenance workers**), blackening of paints, explosion of gases (methane)
- **Prevention and control:** keep sewage fresh and **aerobic**: keep velocity above 0.6 m/s, ventilation, avoid long detention, **air or oxygen injection** into force mains, steep gradients
- **Chemical dosing:** chlorine, hydrogen peroxide, ferrous / ferric salts (precipitate sulphide), lime (raise pH above 9), nitrate, ozone
- **Use of resistant materials:** vitrified clay, HDPE / PVC, **epoxy / PVC lining on concrete**, high alumina cement, additional sacrificial cover, **coat with bitumen**
- Control industrial discharges (acidic waste, high sulphate); regular inspection and cleaning of slime

11. Sewage Pumping

Pumping of sewage is needed when the ground is **flat**, sewers become too deep, or low-lying areas are to be drained, or the effluent must be lifted to treatment works.

- **Pumping stations:** wet well (receives sewage, with **screens** before it) + dry well (pumps and motors) or **submersible pumps** in the wet well
- Pump types: **centrifugal non-clog pumps** with open or semi-open impeller, **screw pumps** (Archimedean) for large flows at low heads, **pneumatic ejectors**, submersible pumps
- **Wet well detention time** 10-30 min (to avoid septicity); pumps start and stop by **float switches**; standby pump provided
- Rising mains (force mains): velocity **1.0 – 2.0 m/s**; air valves and scour valves provided; flushing arrangement
- Design capacity = **peak flow** (2 to 3 x avg); station must have **screening, emergency power supply, alarms, ventilation**

12. Drainage in Buildings and Plumbing Systems

Building drainage conveys waste water and soil from fixtures to the public sewer, safely and hygienically. Design follows the **National Building Code and IS 1742 / IS 2065**.

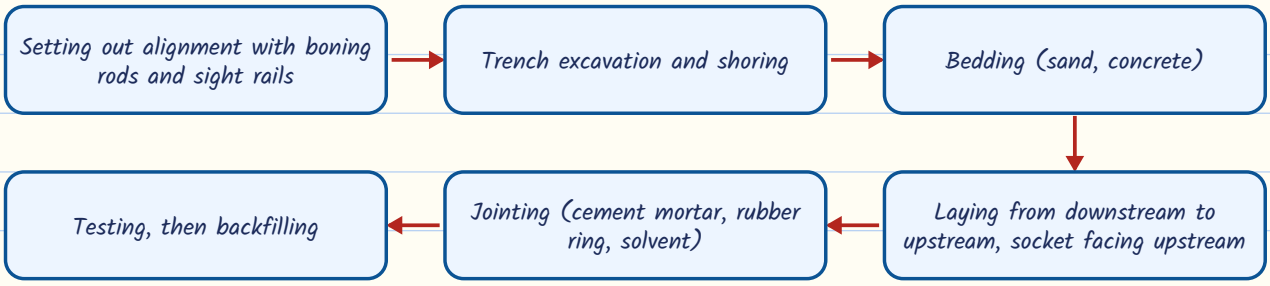
Fixtures and traps

- **Soil pipe** - carries discharge from WCs and urinals; **waste pipe** - from wash basins, baths, sinks, kitchen; **anti-siphonage pipe** (vent) - protects water seals; **vent (stack) pipe** - upto roof, 2 m above roof, **rain water pipes**
- **Traps** - U-shaped bends holding a **water seal** (25-75 mm) to prevent entry of foul gas into the building: P-trap, S-trap, Q-trap, gully trap, intercepting trap (between house drain and sewer), floor trap, grease trap
- **Seal can be lost** by **siphonage, back pressure, evaporation, capillary action** - prevented by venting (anti-siphonage pipes)

System of plumbing	Description
Single stack system	Single pipe for soil and waste; fixtures close to stack; traps with deep seals
One pipe system	Soil and waste in a common stack, **with vents** to each trap
Two pipe system	Separate soil stack and waste stack, with vent pipes; **most hygienic** but costly
Partially ventilated single stack	Reduced vents, for low-rise buildings

- Gradient of house drains 1 in 40 (100 mm) to 1 in 60 (150 mm); **minimum 100 mm diameter**
- **Inspection chambers** at every change; **intercepting trap** at the last chamber; sewage connected to public sewer through a **Y-junction**
- **Rain water pipes** must be separate; **septic tank** in the absence of sewers
- **Testing:** water test (head 1.5 m), smoke test, air test, peppermint test to find leaks

13. Construction, Testing and Maintenance of Sewers



- **Laying** starts at the lower end; the pipes are laid on a **uniform gradient** using a **boning rod and traveller**; trench width = D + 0.3 m each side; **concrete cradle /**

haunch for heavy loads

- **Joints:** cement mortar (1:1), collar, **rubber ring (flexible) joints**, bituminous, lead
- **Testing of sewers: water test** (head of 1.5 m above crown at upstream end, fall in water level checked over 30 min), **air test** (pressure 100 mm water gauge falling by < 25 mm in 5 min), **smoke test**, **mirror and lamp test** for alignment, **ball test** for obstruction
- **Cleaning:** flushing, rodding, **sewer cleaning machines (jetting, bucket machines, hydraulic rodding)**; removal of silt, grease, roots; **record maps** and regular inspection

Sewer gases and safety

- **Hydrogen sulphide** (toxic, 0.1% fatal), **methane** (explosive, 5–15%), **carbon dioxide, carbon monoxide, ammonia**, oxygen deficiency in manholes
- **Safety precautions:** test the atmosphere, **forced ventilation**, no naked flame or smoking, wear **breathing apparatus and safety harness**, never enter a sewer alone, rescue arrangement; **manual scavenging is banned by law**

Example: invert levels of two consecutive sewers

Manhole M1 invert = 100.00 m; sewer M1–M2 length 60 m, gradient 1 in 200, D = 300 mm. Fall = 60/200 = 0.30 m.

Invert of the first sewer at M2 = 99.70 m. If the downstream sewer M2–M3 has D = 450 mm, **match the crown levels**: crown of upstream sewer at M2 = 99.70 + 0.30 = 100.00 m, so the invert of the larger pipe = 100.00 − 0.45 = **99.55 m** (drop of 0.15 m in the manhole).

Remember for exam

* Sewage = 99.9% water; 80% of water supply becomes sewage; per capita BOD 0.08 kg/day
* Min velocity 0.6 m/s, max 3 m/s; max Q at 0.94D, max V at 0.81D
* Rational formula Q = CiA/360; Manning n = 0.013 for sewers
* Corrosion due to H_2S -> H_2SO_4 at crown; control: keep aerobic, ventilation, lining
* Separate system is best for India; manhole spacing 30 m (small sewers)
* Traps maintain water seal to stop sewer gas

Important questions

1. Explain the characteristics and composition of sewage. Define BOD, COD and population equivalent.
2. Compare the separate and combined systems of sewerage.
3. Explain how the quantity of sanitary sewage is estimated.

4. Describe materials used for sewers with their merits and demerits.
5. Explain self-cleansing velocity. Design a circular sewer for a given population.
6. Explain storm runoff estimation by the Rational method.
7. Describe sewer appurtenances (manhole, drop manhole, inverted siphon, catch basin) with sketches.
8. What is sewer corrosion? Explain its mechanism, prevention and control.
9. Explain building drainage, traps and the one-pipe and two-pipe systems of plumbing.

Environmental Engineering

Unit V: Sewage Treatment and Disposal

Syllabus

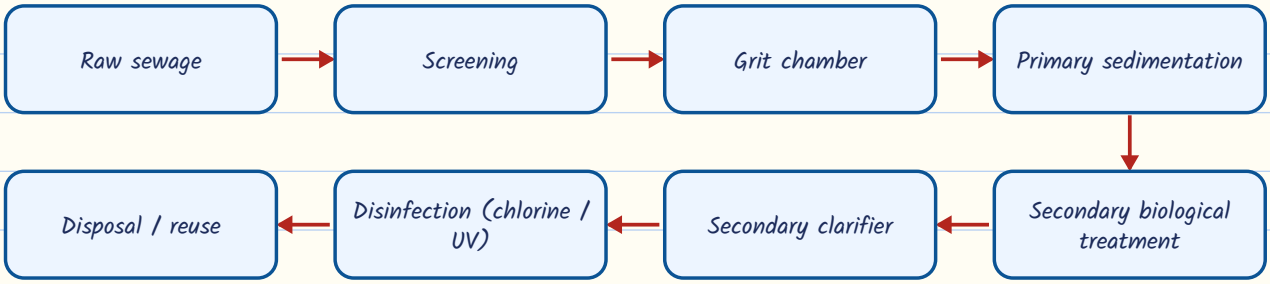
Objectives - Selection of treatment methods - Principles, functions - Activated sludge process and extended aeration systems - Trickling filters - Sequencing batch reactor (SBR) - UASB - Waste stabilization ponds - Other treatment methods - Reclamation and reuse of sewage - Recent advances in sewage treatment - Construction, operation and maintenance aspects - Discharge standards - Sludge treatment - Disposal of sludge.

1. Objectives of Sewage Treatment

- To remove **suspended solids, organic matter (BOD)** and pathogens before disposal
- To **protect receiving waters** (rivers, lakes, sea) from oxygen depletion and pollution
- To prevent **public health hazards** and nuisances (smell, flies)
- To produce effluent suitable for **reuse** (irrigation, industry, ground water recharge)
- To recover **resources**: biogas, manure, treated water, nutrients

Levels of treatment

Level	Units	Removes
Preliminary	Screens, grit chamber, skimming tank, comminutor	Floating matter, grit, oil and grease
Primary	Primary sedimentation tank	Settleable solids; **60% SS, 30-35% BOD**
Secondary (biological)	Activated sludge, trickling filter, ponds, SBR, UASB	Dissolved and colloidal organics; **85-95% BOD**
Tertiary / advanced	Nutrient removal, filtration, chlorination, UV, RO	Nitrogen, phosphorus, pathogens, residual organics



Selection of treatment methods depends on

* Characteristics of the sewage and **required effluent standards**
* **Land availability** (ponds need big area; ASP is compact) and cost of land
* **Capital cost, operating cost and power** needs (trickling filters and ponds use less power)
* Skilled operators, climate (ponds and UASB work best in warm climate), reuse of by-products (biogas)
* Future expansion, variations in flow and load (shock loads)

2. Preliminary Treatment

- **Screens** - coarse (bars 25-50 mm apart), medium (12-25 mm), fine (6-12 mm); manual or mechanical; velocity through bars 0.6-1 m/s; screenings are buried or incinerated
- **Comminutors** - cut the solids to small pieces, so that they can be carried in the flow
- **Grit chamber** - removes inorganic grit (sand, gravel, 0.2 mm, S.G. 2.65) to protect pumps from wear; **horizontal flow velocity = 0.3 m/s**, detention time 30-60 s; types: constant velocity (parabolic channel, proportional weir), **aerated grit chamber**, vortex
- **Skimming tanks** - remove oil and grease by floating them on the surface with air bubbles; detention 3 min
- **Equalisation tank** - balances flow and load variations

3. Primary Sedimentation

- Settles the organic suspended solids to get **raw primary sludge** (3-5% solids)
- Rectangular (L:B = 3-5:1) or circular (diameter 20-40 m, side-water depth 3-4 m); **detention time 1.5 – 2.5 h**; surface loading **25 – 40 $m^3/m^2/day$** (up to 50 at peak); weir loading 125-250 $m^3/m/day$
- Mechanical scrapers move sludge to hoppers; **Imhoff tank** (two-storey: sedimentation + digestion) is for small plants
- Removes 60-70% SS and 30-35% BOD

4. Biological Treatment Principles

Micro-organisms (bacteria, protozoa) consume the dissolved organic matter and use it for **energy and growth**; the cell mass (sludge) is then separated. **Aerobic** processes use oxygen, **anaerobic** work without oxygen (producing CH_4).

Aerobic oxidation

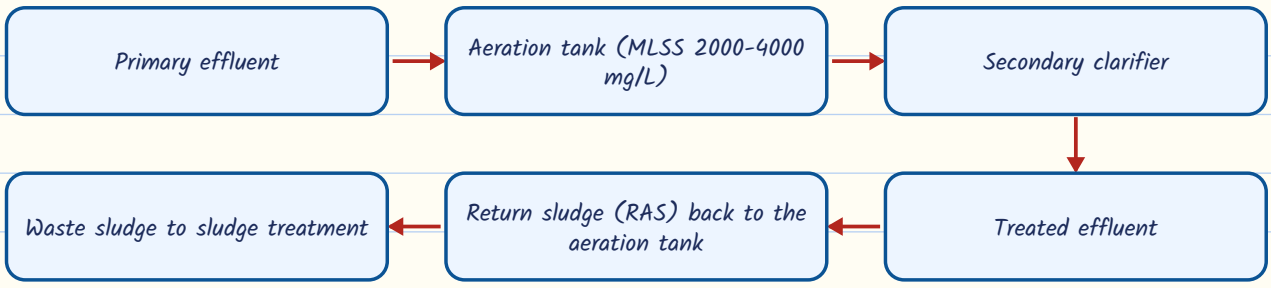
Organic matter + O_2 + bacteria -> CO_2 + H_2O + new cells + energy

Anaerobic: organic matter -> organic acids -> **CH_4 + CO_2** (methanogens)

Growth type	Description
Suspended growth	Micro-organisms in suspension in the aeration tank (activated sludge, ponds, SBR)
Attached growth	Biofilm on a medium: trickling filters, RBC, MBBR
Aerobic	Needs DO of 1.5-2 mg/L
Anaerobic	UASB, digesters, septic tank, Imhoff tank

5. Activated Sludge Process (ASP)

The most widely used **suspended growth aerobic process**. Settled sewage is mixed with a mass of micro-organisms (**activated sludge**) in an **aeration tank**, and air supplied. The mixed liquor flows to a **secondary clarifier** where sludge settles; a part of the **settled sludge is recycled** (return activated sludge, RAS) and the rest wasted (WAS).



Important parameters

Parameter	Definition / typical value
MLSS	Mixed liquor suspended solids; 2000 - 4000 mg/L (conventional)
MLVSS	Volatile fraction (about 70-80% of MLSS)
F/M ratio	Food to micro-organism ratio = Q BOD_0 / (V X) = kg BOD per day / kg MLSS; **0.2 - 0.5 per day** (conventional); extended aeration 0.05-0.15
HRT (theta)	V / Q = 4 - 8 h (conventional)
SRT / MCRT ($theta_c$)	Sludge age = V X / ($Q_w X_r + Q_e X_e$) = 5 - 15 days
SVI	Sludge volume index = (30-min settled volume in mL/L) / (MLSS in g/L); good settling **50 - 150 mL/g**; above 150 indicates bulking
Recycle ratio R	Q_r/Q = X / (X_r - X) = 0.25 - 1.0
Volumetric loading	0.3 - 0.7 kg BOD/m^3/day
Oxygen requirement	About 1 - 1.2 kg O_2 per kg BOD removed
BOD removal efficiency	85 - 95%

Design equations

$V = theta_c\ Y\ Q\ (S_0 - S) / [X\ (1 + k_d\ theta_c)]$ (Y = yield 0.4-0.8, k_d = decay 0.025-0.075 per day)

Sludge produced: $P_x = Y\ Q\ (S_0 - S) / (1 + k_d\ theta_c)$

Efficiency $E = (S_0 - S) / S_0 \times 100$

Example

Q = 5 MLD, BOD_0 = 150 mg/L (after primary), MLSS = 3000 mg/L, F/M = 0.3. BOD load = 5000 x 150 /1000 = 750 kg/d.

V = 750/(0.3 x 3) = 833 m^3 => HRT = 833/5000 x 24 = 4 h. With depth 4 m, area = 208 m^2.

Aeration systems

- **Diffused aeration** - fine bubble diffusers (membrane, ceramic) at bottom, air from blowers; high oxygen transfer efficiency
- **Mechanical (surface) aerators** - low-speed vertical turbines or brush aerators
- Mixing needs and oxygen demand both govern; **DO 1.5 - 2 mg/L** maintained

Modifications of ASP

Modification	Key feature
Conventional (plug flow)	Uniform sewage input at one end; F/M 0.2 – 0.4
Step aeration	Sewage fed in steps along the tank; uniform oxygen demand
Tapered aeration	Air supply decreases along length
Complete mix	Inlet and outlet distributed; resistant to shock loads
Contact stabilisation	Short contact and separate re-aeration of RAS
Extended aeration	Long sludge age, low F/M, see below
Oxidation ditch	Racetrack channel with brush rotors
High-rate ASP	High F/M 1.5 – 5, short HRT 0.5 – 2 h

Operational problems

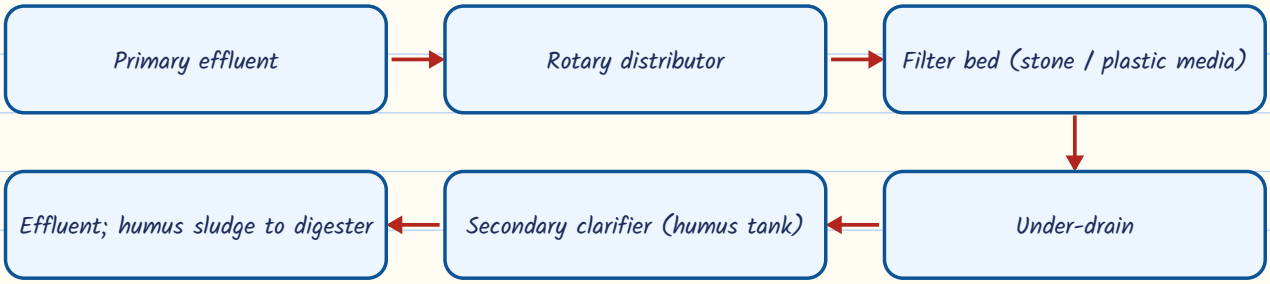
- **Sludge bulking** – filamentous bacteria (Sphaerotilus, Nocardia), SVI above 150, poor settling; remedy: chlorinate return sludge, adjust F/M, increase DO
- **Rising sludge** (denitrification in clarifier) and **foaming**
- **Pin floc**, **toxic shock** loads, **low DO**

6. Extended Aeration Systems

- A variant of ASP where sewage is aerated for a **long time (18 – 36 h)**, with **low F/M (0.05 – 0.15)** and a high sludge age **(20 – 30 days)**
- Micro-organisms start **endogenous respiration** (they consume each other, so sludge is **stabilised aerobically**); the excess sludge is small and fairly stable, needs no digestion
- **No primary settling** needed (raw screened sewage after grit); MLSS 3000-6000 mg/L; aeration tank volume large; **oxidation ditch** is an example
- Suited to **small communities, hotels, hostels, packaged plants**; simple operation; **BOD removal 90-95%**, nitrification occurs
- Disadvantages: **high power cost (aeration)**, big tank, high MLSS in the effluent solids

7. Trickling Filters

A bed of **coarse media (stones, plastic, slag)** over which settled sewage is sprayed or distributed intermittently. A **biofilm (zoogleal film)** grows on the surface of the media; as sewage trickles down, the organics are absorbed and oxidised, with **air flowing naturally** through voids.



- **Components:** circular tank, **rotary distributor** (hydraulic arms), **filter medium** (stone 40-80 mm, depth 1.8-2.5 m; plastic modules 4-12 m), **under-drainage** (with ventilation), **humus tank** (secondary settling)
- Sloughing of the biofilm (humus) is removed in the humus tank

Parameter	Low (standard) rate	High rate
Hydraulic loading	1 - 4 $m^3/m^2/day$	10 - 40 $m^3/m^2/day$
Organic loading	80 - 320 g $BOD/m^3/day$ (0.08-0.32 kg)	320 - 1000 g $BOD/m^3/day$
Depth	1.8 - 3 m	1 - 2 m (stone), up to 12 m plastic
Recirculation	None	**1 - 4 times**
BOD removal	80 - 90%	65 - 80%
Effluent	Nitrified, stable	Not nitrified
Problems	Filter flies (Psychoda), odour	Ponding, odour

NRC formula (high rate, single stage)

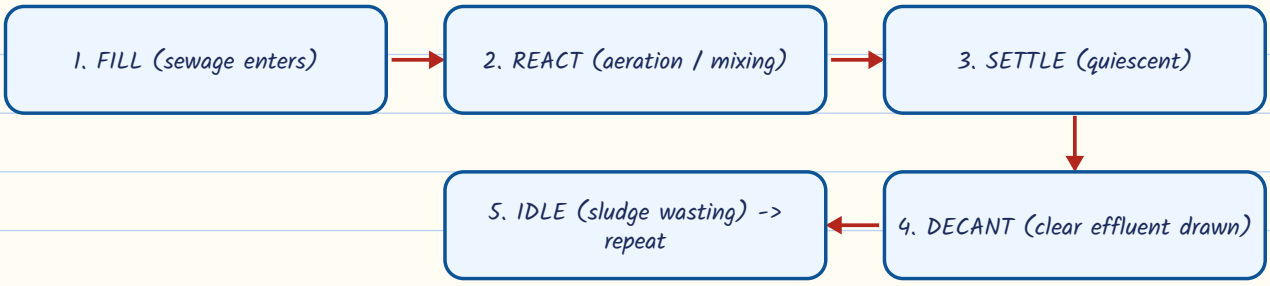
$E = 100 / (1 + 0.4432\ sqrt(W / (V F)))$ (W = BOD load in kg/day, V = volume in m^3, F = recirculation factor = $(1 + R)/(1 + 0.1R)^2$)

- **Merits:** low power, simple, resists shock loads, low sludge; **demerits:** big land, odour and flies, clogging, lower efficiency in cold

- **Related attached-growth processes: Rotating biological contactor (RBC)** - discs rotating partly in sewage; **Biotowers**; **Moving bed bio-reactors (MBBR)**; **Submerged aerated filters**

8. Sequencing Batch Reactor (SBR)

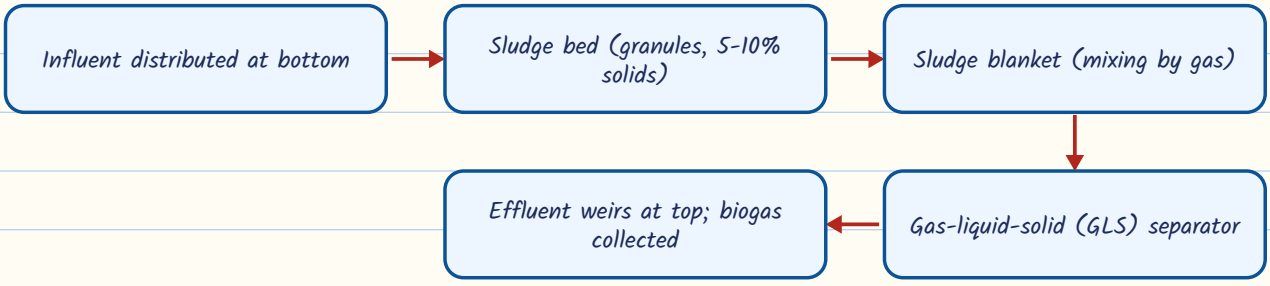
The SBR is a **fill-and-draw activated sludge process** in which all steps - aeration and clarification - take place in **a single tank** in a time sequence. No separate clarifier and no sludge return pump are needed.



- Cycle 4 - 8 h (fill 1-2 h, react 2-4 h, settle 0.5-1 h, decant 0.5-1 h); **at least two tanks** work alternately for continuous inflow
- **Decanter** (floating or fixed) removes treated water; **PLC / SCADA** controls the cycle
- **Merits:** compact, flexible, good **nutrient removal (nitrogen and phosphorus)** by varying aerobic and anoxic phases, no sludge bulking issues, no secondary clarifier
- **Demerits:** needs **automatic control and skilled maintenance**, peak flow storage, decanter issues. Variants: **CASS, ICEAS, MBBR-SBR**

9. UASB (Upflow Anaerobic Sludge Blanket) Reactor

An **anaerobic** treatment unit in which sewage flows **upward** through a **dense blanket of granular sludge** at the reactor bottom. Anaerobic bacteria break the organics into **biogas (60-70% methane)**. Developed in the Netherlands (Lettinga); widely used in India (Karnal, Kanpur, Mirzapur under the Ganga Action Plan).



- **Design parameters:** HRT **6 – 8 h** (domestic sewage), upflow velocity **0.5 – 1.0 m/h**, height 4 – 6 m, **OLR 1 – 3 kg COD/m^3/day**, temperature 20-35 degrees C
- **BOD removal 60 – 80%**; produces **biogas (energy)**, and **little sludge (well stabilised)**; low power; low cost; the effluent needs **post-treatment** (polishing pond, ASP, or aerated lagoon) to meet standards
- **Demerits:** slow start-up (months, seeding needed), odour (H_2S), sensitive to toxic shocks and low temperature, effluent with high COD and pathogens

10. Waste Stabilization Ponds (WSP)

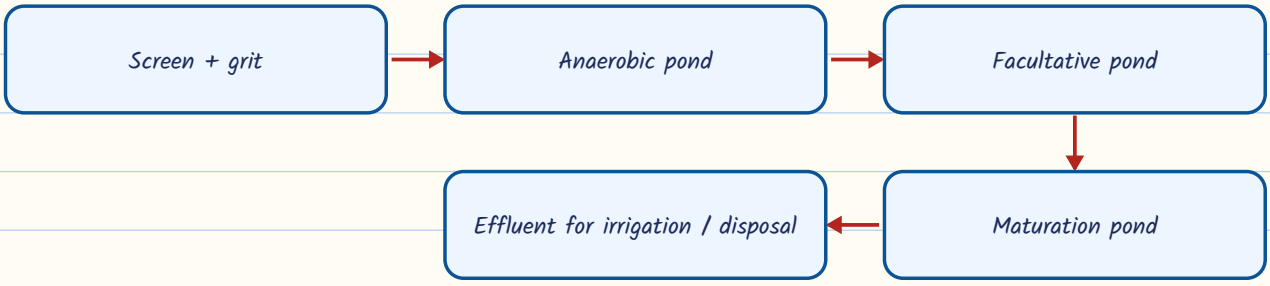
Shallow earthen basins in which sewage is treated by **natural processes** (sunlight, algae, bacteria, wind). Algae produce oxygen by **photosynthesis** for bacteria that oxidise organic matter; this **symbiosis** is the principle.

Algal-bacterial symbiosis

Organic matter + O_2 (from algae) --bacteria--> CO_2 + NH_3 + PO_4 + new cells

CO_2 + nutrients + sunlight --algae--> new algae + O_2

Pond	Depth	Detention	Features
Anaerobic	2.5 – 5 m	2 – 5 d	Heavy BOD (OLR up to 400 g/m^3/d), sludge digests, 50-60% BOD removal, odour
Facultative	1.0 – 1.5 m	5 – 30 d	Aerobic top, anaerobic bottom; most common; BOD loading 100-400 kg/ha/d
Maturation (aerobic)	0.8 – 1.2 m	5 – 10 d	Removes **pathogens (bacteria)**, N, with UV; low BOD feed
Aerated lagoon	2 – 4 m	3 – 10 d	Mechanical aerators, higher loads



- **Merits:** cheap, **no power**, simple O and M, high pathogen removal; **demerits: large land**, odour in anaerobic ponds, mosquitoes, algal in effluent (SS), seepage
- **Design:** area from BOD loading (kg/ha/day) = BOD x Q / area; e.g., 1 MLD at 200 mg/L is 200 kg/d; at 200 kg/ha/d needs 1 ha

- **Fish and aquatic (duckweed / water hyacinth) ponds** are also used for polishing

11. Other Treatment Methods

Method	Notes
Septic tank + soak pit	On-site, for houses; detention 24 h; digestion of settled sludge, 30-50% BOD removal; effluent to soak pit or dispersion trench
Imhoff tank	Settling above and digestion below
Oxidation ditch	Extended aeration, racetrack; brush rotors
Rotating biological contactor (RBC)	Discs 40% submerged, 1-2 rpm
Anaerobic filter, Anaerobic baffled reactor	Small plants
Constructed wetlands / root zone	Reeds and gravel; decentralised, natural
Membrane bio-reactor (MBR)	ASP + membrane filtration, no clarifier; excellent effluent
Moving bed biofilm reactor (MBBR)	Plastic carriers in suspended aeration tank; compact
Fluidised bed reactor	Biofilm on sand carriers

12. Reclamation and Reuse of Sewage

With shortage of fresh water, treated sewage is a source of water. **Reclamation** = treating the wastewater to a quality fit for a beneficial use; **reuse** is the use.

Reuse	Required treatment
Agriculture / irrigation (non-food crops, orchards, with food crops need higher quality)	Secondary + disinfection; sewage farming
Industrial (cooling water, boilers, process)	Secondary + filtration + softening
Urban non-potable (toilet flushing, gardens, fire fighting, road washing, car wash)	Secondary + filtration + disinfection
Ground water recharge (injection, spreading basins)	Tertiary + RO
Recreational lakes, aquaculture (fish ponds)	Secondary + maturation / disinfection
Indirect / direct potable reuse	Advanced: MF - RO - UV - AOP (e.g., NEWater Singapore, Windhoek)

- **Risks:** pathogens, heavy metals, salts, emerging contaminants (pharmaceuticals); need strict **monitoring and standards**, public acceptance

- **Benefits:** water saving, nutrients (N, P) for crops, less pollution load, drought-proof supply, reduced fresh water demand
- Policy: **National Water Policy** and Urban Wastewater reuse policies encourage use of treated sewage by industries and for irrigation

13. Recent Advances in Sewage Treatment

- **Membrane bioreactors (MBR)** and **Anaerobic MBR**
- **Aerobic granular sludge (Nereda)** - compact, nutrient removal, no clarifier
- **Anammox (anaerobic ammonium oxidation)** - cheaper nitrogen removal; **enhanced biological phosphorus removal (EBPR); A2O / Bardenpho / modified UCT processes**
- **Advanced oxidation processes (AOP):** ozone, UV/H_2O_2 for micropollutants
- **Constructed wetlands, phytoremediation, decentralised systems (DEWATS)**
- **Microbial fuel cells, energy-positive plants** (biogas CHP), **resource recovery** (struvite, phosphorus)
- **Smart operation:** online sensors, **SCADA / AI-based control**; **Hybrid Annuity Model (HAM)** plants under **Namami Gange**
- **Nano-filtration, forward osmosis, electrocoagulation**, and **solar-powered treatment**

14. Construction, Operation and Maintenance Aspects

- **Construction:** units on **hydraulic gradient**, preferably gravity flow, in water-tight RCC; **bypass** and flow measurement (Parshall flume), standby units; plant located **downstream and away** from the town (buffer zone), with flood protection and room for expansion
- **Operation:** monitor flow, DO, MLSS, SVI, F/M, pH, BOD, COD, SS, residual chlorine, and keep **log sheets**; control sludge wasting and recycle; adjust air supply
- **Maintenance:** clean screens daily, remove grit, lubricate gearboxes and scrapers, clean diffusers, repaint, desludge ponds, calibration; **preventive maintenance schedule**, spare parts
- **Safety and health:** toxic gases (H_2S, CH_4), confined spaces, pathogens (vaccination, PPE), **odour control** (bio-filters, covering); **trained staff and laboratory**

15. Discharge Standards (Effluent Standards)

Treated sewage must meet standards of the **Central Pollution Control Board (CPCB)** under the **Environment (Protection) Act, 1986** (and **Water Act, 1974**), before discharging to inland surface water, land for irrigation or marine areas.

Parameter	Inland surface water	Land for irrigation
pH	5.5 - 9.0	5.5 - 9.0
BOD_5 at 20 degrees C	30 mg/L (STPs: 10-20 mg/L in new NGT standards)	100 mg/L
COD	250 mg/L (new STP norms: 50-100)	—
Total suspended solids	100 mg/L (STP: 20-50)	200 mg/L
Oil and grease	10 mg/L	10 mg/L
Ammonical nitrogen	50 mg/L (STP: 5)	—
Total nitrogen	—(STP: 10)	—
Fecal coliform	Less than 1000 MPN/100 mL (desirable 230)	—
Temperature	Shall not exceed 5 degrees C above receiving water	—

Note: The **National Green Tribunal (2019)** and the **CPCB (2017)** have set stricter norms for STPs: BOD below 10 mg/L, TSS 20, FC 100 MPN/100 mL in sensitive areas (values vary by capacity and location; check the latest notification).

16. Sludge Treatment

Sludge is a mixture of solids and water (**98–99% water** in primary sludge, 99–99.5% in activated sludge). Its treatment aims to reduce volume, stabilise organics, destroy pathogens and make it fit for disposal.



Sludge volume and moisture

Relation

$V_1/V_2 = (100 - P_2) / (100 - P_1)$ (P = % moisture)

Example: 100 m^3 sludge at 98% moisture reduces to 100 x 2/5 = **40 m^3** at 95%.

Sludge Volume Index (SVI) - settling characteristic of activated sludge

Anaerobic digestion

- Stabilises sludge in closed tanks without oxygen; **temperature 35 degrees C (mesophilic)**, retention 20-30 days (high rate 10-15 d); pH 6.8-7.2
- **Stages:** (1) **hydrolysis**, (2) **acid formation (acidogenesis)**, (3) **methanogenesis (methane formation)**
- **Biogas** - 60-65% CH_4, 30-35% CO_2, traces of H_2S; **0.8 - 1.1 m^3 gas per kg volatile solids destroyed**; used for heating the digester or for power generation
- **Types:** standard rate (unheated, no mixing), **high-rate (heated, mixed)**, **two-stage** (primary + secondary)
- **Digester gas** is collected in gasholder; **digested sludge** is dark, inoffensive and with 50-60% volatile reduction
- **Operational problems:** foaming, scum formation, pH drop (sour digester), toxicity

Conditioning and dewatering

- **Conditioning:** chemical (alum, ferric chloride, lime, **polyelectrolytes**), heat treatment, elutriation
- **Sludge drying beds** - open sand beds (15-30 cm sand + 20-30 cm gravel), 7-15 days drying, solid content 30-40%, cheapest; needs area (0.1 - 0.25 m^2/person); covered in monsoon
- **Mechanical: centrifuge, belt filter press, filter press, vacuum filter** - 15-35% solids, used in big plants
- **Thermal drying and incineration**, **composting**, **vermi-stabilisation**

17. Disposal of Sludge

Disposal method	Notes
Land application / manure	Digested, dried sludge as soil conditioner (N, P, organic matter); free from pathogens and heavy metals as per norms
Landfill (sanitary)	Dried sludge, with solid waste; control leachate and gas
Lagooning	Sludge in earthen lagoons, natural stabilisation
Incineration	Reduces volume to ash by burning at 800-1000 degrees C; costly, air pollution control
Composting	With solid waste, high-quality compost
Use as fuel / brick making / cement kiln (co-processing)	Resource recovery
Disposal into sea	Banned or restricted

Remember for exam

* Primary removes 30-35% BOD; secondary 85-95%
* ASP: F/M 0.2-0.5, MLSS 2000-4000, SVI 50-150, sludge age 5-15 d; extended aeration F/M 0.05-0.15, 18-36 h
* Trickling filter: biofilm, rotary distributor, high rate with recirculation
* SBR: Fill - React - Settle - Decant - Idle, one tank
* UASB: anaerobic, 6-8 h, biogas 60-70% CH_4, needs post-treatment
* WSP: algal-bacterial symbiosis; anaerobic - facultative - maturation
* CPCB standard BOD 30 mg/L (inland water); digestion at 35 degrees C produces biogas

Important questions

1. Explain the objectives and the selection of sewage treatment methods. Draw the flow diagram.
2. Describe the activated sludge process. Explain F/M ratio, MLSS, SVI and sludge age; discuss sludge bulking.
3. Explain the extended aeration system and compare it with conventional ASP.
4. Describe the working of a trickling filter. Differentiate low-rate and high-rate filters.
5. Explain the sequencing batch reactor and the UASB reactor with neat sketches.
6. Describe waste stabilisation ponds and the algal-bacterial symbiosis.
7. Discuss reclamation and reuse of sewage and the recent advances in treatment.
8. Explain sludge digestion, conditioning, dewatering and disposal of sludge.

9. Write notes on discharge standards and O and M aspects of a sewage treatment plant.